

Markov Chains

A Comprehensive Study Guide

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1 Introduction

1.1 Historical Background

Markov chains are named after the Russian mathematician **Andrey Markov** (1856-1922), who first studied them in the early 20th century. In 1906, Markov published his groundbreaking paper introducing what we now call Markov chains. His work was motivated by a disagreement with Pavel Nekrasov, who claimed that independence was necessary for the law of large numbers to hold.

Markov demonstrated that the weak law of large numbers could hold for dependent random variables, provided they satisfied certain conditions—what we now call the Markov property. He famously applied his theory to analyze the distribution of vowels and consonants in Alexander Pushkin’s novel *Eugene Onegin*.

Year	Milestone
1906	Markov publishes first paper on Markov chains
1912	Henri Poincaré studies Markov chains on finite groups
1928	Maurice Fréchet’s detailed study
1931	Andrey Kolmogorov develops continuous-time theory
1948	Claude Shannon applies to information theory
Present	Widespread applications in ML, finance, biology

1.2 What is a Markov Chain?

A **Markov chain** is a stochastic (random) process that satisfies the **Markov property**: the probability of transitioning to the next state depends only on the current state and not on the sequence of states that preceded it. This “memoryless” property is the defining characteristic of Markov chains.

The Markov Property (Informal): “The future is independent of the past, given the present.”

Mathematical Definition:

$$P(X_{n+1} = x \mid X_0 = x_0, X_1 = x_1, \dots, X_n = x_n) = P(X_{n+1} = x \mid X_n = x_n)$$

1.3 Key Terminology

- **State Space (S):** The set of all possible states the system can be in
- **Transition:** A change from one state to another
- **Transition Probability:** The probability of moving from state i to state j
- **Transition Matrix (P):** A matrix containing all transition probabilities
- **Initial Distribution:** The probability distribution over states at time $t = 0$

2 Types of Markov Chains

2.1 Classification by Time Parameter

Type	Description
Discrete-Time Markov Chain (DTMC)	State changes occur at discrete time steps (e.g., $t = 0, 1, 2, \dots$)
Continuous-Time Markov Chain (CTMC)	State changes can occur at any time (continuous parameter)

2.2 Classification by State Space

Type	Description
Finite State Space	Countable and finite number of states
Countably Infinite	States can be enumerated (e.g., $\mathbb{N}$)
Continuous State Space	Uncountable state space (e.g., $\mathbb{R}$)

3 Mathematical Foundations

3.1 Formal Definition: Discrete-Time Markov Chain

A **discrete-time Markov chain** is a sequence of random variables $X_0, X_1, X_2, \dots$ with the Markov property:

$$P(X_{n+1} = j \mid X_n = i, X_{n-1} = i_{n-1}, \dots, X_0 = i_0) = P(X_{n+1} = j \mid X_n = i)$$

where both conditional probabilities are well-defined.

3.1.1 Time-Homogeneous Markov Chains

A Markov chain is **time-homogeneous** (or **stationary**) if the transition probabilities are independent of time:

$$P(X_{n+1} = j \mid X_n = i) = P(X_1 = j \mid X_0 = i) = p_{ij}$$

for all $n \geq 0$.

3.2 Transition Matrix

For a finite state space $S = \{1, 2, \dots, N\}$, the **transition matrix** P is an $N \times N$ matrix where:

$$P = \begin{pmatrix} p_{11} & p_{12} & \dots & p_{1N} \\ p_{21} & p_{22} & \dots & p_{2N} \\ \vdots & \vdots & \ddots & \vdots \\ p_{N1} & p_{N2} & \dots & p_{NN} \end{pmatrix}$$

3.2.1 Properties of Transition Matrix

1. **Non-negativity:** $p_{ij} \geq 0$ for all i, j
2. **Row Stochastic:** Each row sums to 1:

$$\sum_{j=1}^N p_{ij} = 1 \quad \text{for all } i$$

3. **n-Step Transition Probability:**

$$P^n = \underbrace{P \cdot P \cdot \dots \cdot P}_{n \text{ times}}$$

4. **Chapman-Kolmogorov Equation:**

$$p_{ij}^{(n+m)} = \sum_{k=1}^N p_{ik}^{(n)} p_{kj}^{(m)}$$

3.2.2 Detailed Derivation: n-Step Transition Probabilities

The n -step transition probability $p_{ij}^{(n)}$ is the probability of being in state j after exactly n steps, starting from state i .

Base Cases:

- **n = 0:** $p_{ij}^{(0)} = \begin{cases} 1 & \text{if } i=j \\ 0 & \text{if } i \neq j \end{cases}$ (identity matrix)
- **n = 1:** $p_{ij}^{(1)} = p_{ij}$ (one-step transition)

Recursive Formula:

$$p_{ij}^{(n+1)} = \sum_{k=1}^N p_{ik}^{(n)} p_{kj}$$

Proof by Induction:

Base case (n=1): Already have $p_{ij}^{(1)} = p_{ij}$ ✓

Inductive step: Assume P^n gives n -step probabilities. Show $P^{n+1} = P^n \cdot P$.

By law of total probability:

$$\begin{aligned} p_{ij}^{(n+1)} &= P(X_{n+1} = j \mid X_0 = i) \\ &= \sum_{k=1}^N P(X_{n+1} = j, X_n = k \mid X_0 = i) \\ &= \sum_{k=1}^N P(X_{n+1} = j \mid X_n = k, X_0 = i) \cdot P(X_n = k \mid X_0 = i) \end{aligned}$$

By Markov property:

$$\begin{aligned} &= \sum_{k=1}^N P(X_{n+1} = j \mid X_n = k) \cdot P(X_n = k \mid X_0 = i) \\ &= \sum_{k=1}^N p_{kj} \cdot p_{ik}^{(n)} = \sum_{k=1}^N p_{ik}^{(n)} p_{kj} \end{aligned}$$

This is exactly the (i, j) entry of the matrix product $P^n \cdot P = P^{n+1}$. □

Matrix Power Interpretation:

Matrix Power	Meaning	Calculation
P^0	Identity (no steps)	I
P^1	One-step transitions	P
P^2	Two-step transitions	$P \cdot P$
P^3	Three-step transitions	$P^2 \cdot P$
P^n	n -step transitions	$P^{n-1} \cdot P$

Example: Computing P^3 for 2×2 Matrix

Let $P = \begin{pmatrix} 0.7 & 0.3 \\ 0.4 & 0.6 \end{pmatrix}$

Step 1: Compute P^2

$$P^2 = \begin{pmatrix} 0.7 & 0.3 \\ 0.4 & 0.6 \end{pmatrix} \begin{pmatrix} 0.7 & 0.3 \\ 0.4 & 0.6 \end{pmatrix}$$

Entry (1,1): $0.7 \times 0.7 + 0.3 \times 0.4 = 0.49 + 0.12 = 0.61$ Entry (1,2): $0.7 \times 0.3 + 0.3 \times 0.6 = 0.21 + 0.18 = 0.39$ Entry (2,1): $0.4 \times 0.7 + 0.6 \times 0.4 = 0.28 + 0.24 = 0.52$ Entry (2,2): $0.4 \times 0.3 + 0.6 \times 0.6 = 0.12 + 0.36 = 0.48$

$$P^2 = \begin{pmatrix} 0.61 & 0.39 \\ 0.52 & 0.48 \end{pmatrix}$$

Step 2: Compute $P^3 = P^2 \cdot P$

$$P^3 = \begin{pmatrix} 0.61 & 0.39 \\ 0.52 & 0.48 \end{pmatrix} \begin{pmatrix} 0.7 & 0.3 \\ 0.4 & 0.6 \end{pmatrix}$$

Entry (1, 1): $0.61 \times 0.7 + 0.39 \times 0.4 = 0.427 + 0.156 = 0.583$ Entry (1, 2): $0.61 \times 0.3 + 0.39 \times 0.6 = 0.183 + 0.234 = 0.417$ Entry (2, 1): $0.52 \times 0.7 + 0.48 \times 0.4 = 0.364 + 0.192 = 0.556$ Entry (2, 2): $0.52 \times 0.3 + 0.48 \times 0.6 = 0.156 + 0.288 = 0.444$

$$P^3 = \begin{pmatrix} 0.583 & 0.417 \\ 0.556 & 0.444 \end{pmatrix}$$

Interpretation Table:

Steps	$p_{11}^{(n)}$	$p_{12}^{(n)}$	$p_{21}^{(n)}$	$p_{22}^{(n)}$
0	1.000	0.000	0.000	1.000
1	0.700	0.300	0.400	0.600
2	0.610	0.390	0.520	0.480
3	0.583	0.417	0.556	0.444
4	0.575	0.425	0.567	0.433
∞	0.571	0.429	0.571	0.429

Notice convergence to stationary distribution!=== Chapman-Kolmogorov Equation: Detailed Derivation

The Chapman-Kolmogorov equation is fundamental to Markov chain theory. It states that the probability of going from state i to state j in $n + m$ steps equals the sum over all intermediate states k of going from i to k in n steps, then from k to j in m steps.

Theorem: For a Markov chain with transition probabilities p_{ij} :

$$p_{ij}^{(n+m)} = \sum_{k \in S} p_{ik}^{(n)} p_{kj}^{(m)}$$

Proof:

By definition:

$$p_{ij}^{(n+m)} = P(X_{n+m} = j \mid X_0 = i)$$

Using the law of total probability, we condition on the state at time n :

$$p_{ij}^{(n+m)} = \sum_{k \in S} P(X_{n+m} = j, X_n = k \mid X_0 = i)$$

By conditional probability:

$$= \sum_{k \in S} P(X_{n+m} = j \mid X_n = k, X_0 = i) \cdot P(X_n = k \mid X_0 = i)$$

Key step - Apply Markov property:

$$P(X_{n+m} = j \mid X_n = k, X_0 = i) = P(X_{n+m} = j \mid X_n = k)$$

The future state X_{n+m} depends only on $X_n = k$, not on how we got to k .

Therefore:

$$\begin{aligned} p_{ij}^{(n+m)} &= \sum_{k \in S} P(X_{n+m} = j \mid X_n = k) \cdot P(X_n = k \mid X_0 = i) \\ &= \sum_{k \in S} P(X_m = j \mid X_0 = k) \cdot P(X_n = k \mid X_0 = i) \\ &= \sum_{k \in S} p_{kj}^{(m)} \cdot p_{ik}^{(n)} \\ &= \sum_{k \in S} p_{ik}^{(n)} p_{kj}^{(m)} \end{aligned}$$

□

Matrix Interpretation:

In matrix form: $P^{(n+m)} = P^{(n)} \cdot P^{(m)}$

This is just matrix multiplication! The (i, j) entry is:

$$(P^{(n)} \cdot P^{(m)})_{ij} = \sum_k (P^{(n)})_{ik} (P^{(m)})_{kj} = \sum_k p_{ik}^{(n)} p_{kj}^{(m)}$$

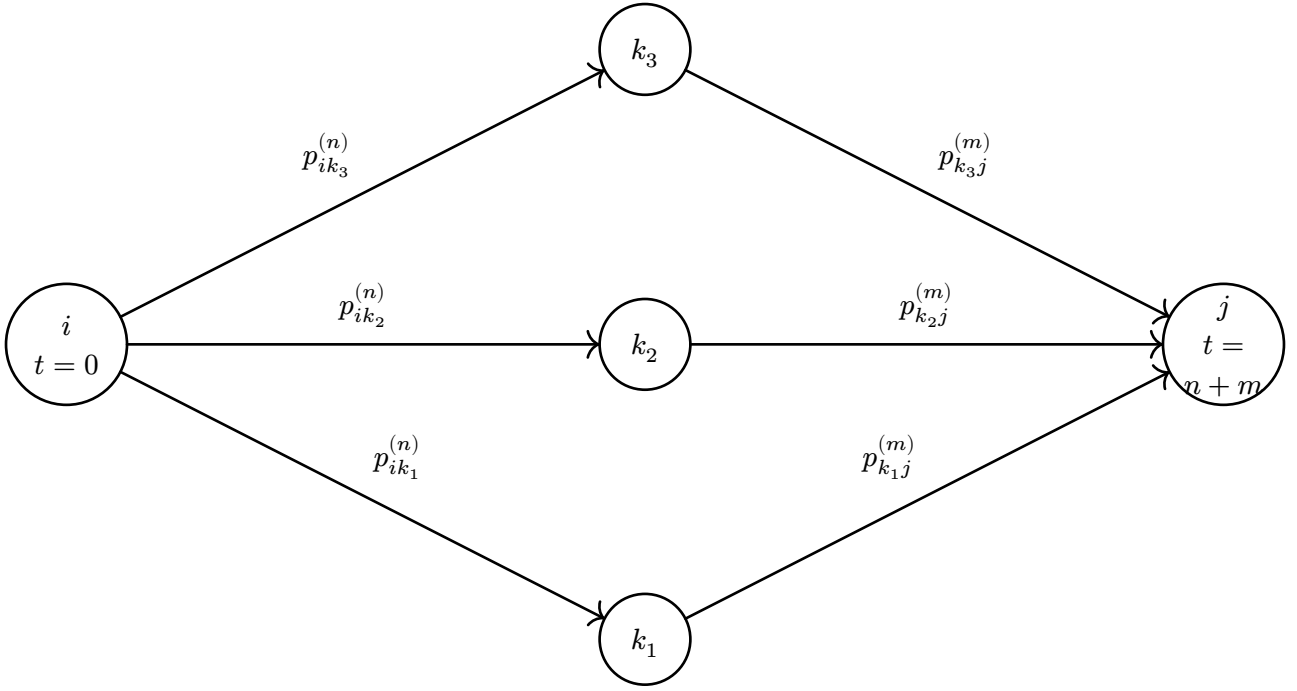
Example Application:

For the weather model with $P = \begin{pmatrix} 0.7 & 0.3 \\ 0.5 & 0.5 \end{pmatrix}$, verify Chapman-Kolmogorov for $n = 1, m = 1$:

$$\begin{aligned} p_{SS}^{(2)} &= \sum_{k \in \{S, R\}} p_{Sk}^{(1)} p_{kS}^{(1)} \\ &= p_{SS}^{(1)} p_{SS}^{(1)} + p_{SR}^{(1)} p_{RS}^{(1)} \\ &= (0.7)(0.7) + (0.3)(0.5) \\ &= 0.49 + 0.15 = 0.64 \end{aligned}$$

✓

Graphical Representation:



The total probability is the sum over all possible intermediate states:

$$p_{ij}^{(n+m)} = p_{ik_1}^{(n)} p_{k_1j}^{(m)} + p_{ik_2}^{(n)} p_{k_2j}^{(m)} + p_{ik_3}^{(n)} p_{k_3j}^{(m)} + \dots$$

3.3 Initial Distribution

The **initial distribution** $\pi^{(0)}$ is a row vector specifying the probability of starting in each state:

$$\pi^{(0)} = [\pi_1^{(0)}, \pi_2^{(0)}, \dots, \pi_N^{(0)}]$$

where:

- $\pi_i^{(0)} = P(X_0 = i)$
- $\sum_{i=1}^N \pi_i^{(0)} = 1$

The distribution at time n is:

$$\pi^{(n)} = \pi^{(0)} P^n$$

4 Examples and Applications

4.1 How to Construct Markov Chain State Machines

Creating a state machine diagram is essential for visualizing Markov chains. Follow this systematic approach:

4.1.1 Step-by-Step Construction Process

Step 1: Identify the State Space

Determine all possible states the system can occupy.

Questions to ask:

- What are the distinct configurations/conditions?
- Are states discrete or continuous?
- Is the state space finite or infinite?

Example: Weather model $\rightarrow$ States: {Sunny, Rainy}

Step 2: Define Transition Probabilities

For each state i , determine:

- Which states j can be reached in one step?
- What is p_{ij} for each reachable state?

Constraint: $\sum_j p_{ij} = 1$ for all i

Step 3: Draw the State Diagram

Components:

- **Nodes (circles):** Represent states
- **Directed edges (arrows):** Represent transitions
- **Edge labels:** Transition probabilities
- **Self-loops:** Transitions from state to itself

Step 4: Construct Transition Matrix

Organize probabilities in matrix form where:

- Rows = current state
- Columns = next state
- Entry $(i, j) = p_{ij}$

Step 5: Verify

Check that:

1. All row sums equal 1
2. All entries are non-negative
3. Diagram matches matrix

4.1.2 Detailed Example: Card Game

Problem: A player has 3 cards. Each round:

- If holding 1 card: gain a card with prob 0.6, lose it with prob 0.4
- If holding 2 cards: gain one (prob 0.5), lose one (prob 0.3), stay same (prob 0.2)
- If holding 3 cards: lose one with prob 0.7, keep all with prob 0.3
- If holding 0 cards: game over (absorbing)

Step 1: State Space

$$S = \{0, 1, 2, 3\}$$

States represent number of cards held.

Step 2: Transition Probabilities

From state 0 (no cards):

- $p_{00} = 1$ (absorbing)

From state 1 (one card):

- $p_{10} = 0.4$ (lose the card)
- $p_{12} = 0.6$ (gain a card)

From state 2 (two cards):

- $p_{21} = 0.3$ (lose one)
- $p_{22} = 0.2$ (stay same)
- $p_{23} = 0.5$ (gain one)

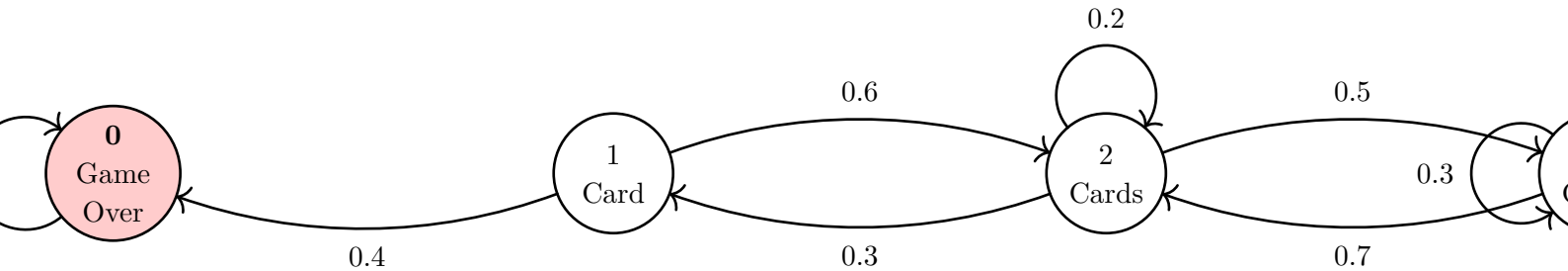
From state 3 (three cards):

- $p_{32} = 0.7$ (lose one)
- $p_{33} = 0.3$ (keep all)

Verification:

- State 0: $1.0 = 1$ ✓
- State 1: $0.4 + 0.6 = 1$ ✓
- State 2: $0.3 + 0.2 + 0.5 = 1$ ✓
- State 3: $0.7 + 0.3 = 1$ ✓

Step 3: State Diagram



Step 4: Transition Matrix

$$P = \begin{pmatrix} 1 & 0 & 0 & 0 \\ 0.4 & 0 & 0.6 & 0 \\ 0 & 0.3 & 0.2 & 0.5 \\ 0 & 0 & 0.7 & 0.3 \end{pmatrix}$$

From To	0	1	2	3
0	1.0	0	0	0
1	0.4	0	0.6	0
2	0	0.3	0.2	0.5
3	0	0	0.7	0.3

Step 5: Analysis

Properties:

- State 0 is absorbing (once game over, stays over)
- States 1, 2, 3 are transient (eventually reach 0)
- Chain is not irreducible (can't leave state 0)
- States 1, 2, 3 are aperiodic (have different periods)

Key Questions:

1. Probability of game ending from state 2?
2. Expected number of rounds from state 3?

These can be answered using absorbing chain theory!

4.1.3 Common State Machine Patterns

Pattern 1: Cyclic Chain States form a cycle: $0 \rightarrow 1 \rightarrow 2 \rightarrow \dots \rightarrow N \rightarrow 0$

Used for: Periodic processes, round-robin scheduling

Pattern 2: Birth-Death Chain Can only move to adjacent states: $i \rightarrow i \pm 1$

Used for: Queues, population dynamics

Pattern 3: Random Walk Symmetric movement: $P(i \rightarrow i + 1) = P(i \rightarrow i - 1) = 0.5$

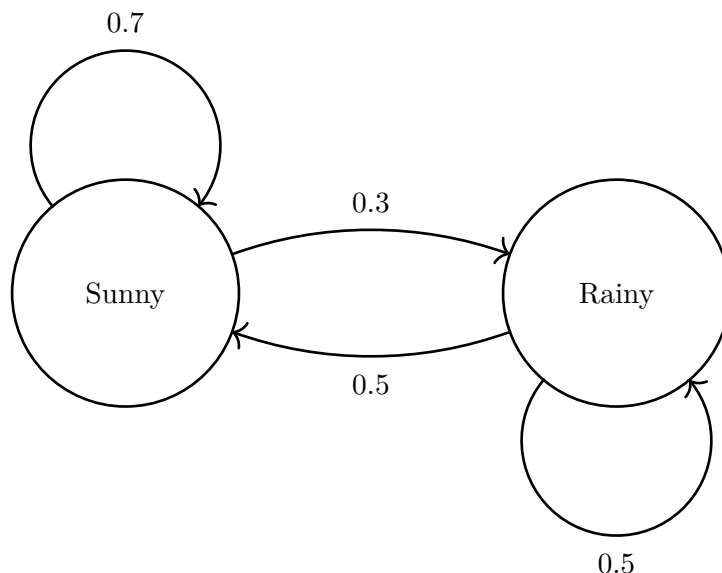
Used for: Brownian motion, stock prices

Pattern 4: Absorbing Chain One or more absorbing states with transient states leading to them

Used for: Games, bankruptcy models, machine failure

4.2 Example 1: Two-State Weather Model

Consider a simplified weather model with two states: Sunny (S) and Rainy (R).



Interpretation:

- If today is sunny, there's a 70% chance tomorrow will be sunny
- If today is rainy, there's a 50% chance tomorrow will be rainy

Transition Matrix:

$$P = \begin{pmatrix} 0.7 & 0.3 \\ 0.5 & 0.5 \end{pmatrix}$$

4.2.1 Computing Future Probabilities

If today is sunny, what's the probability it will be sunny in 2 days?

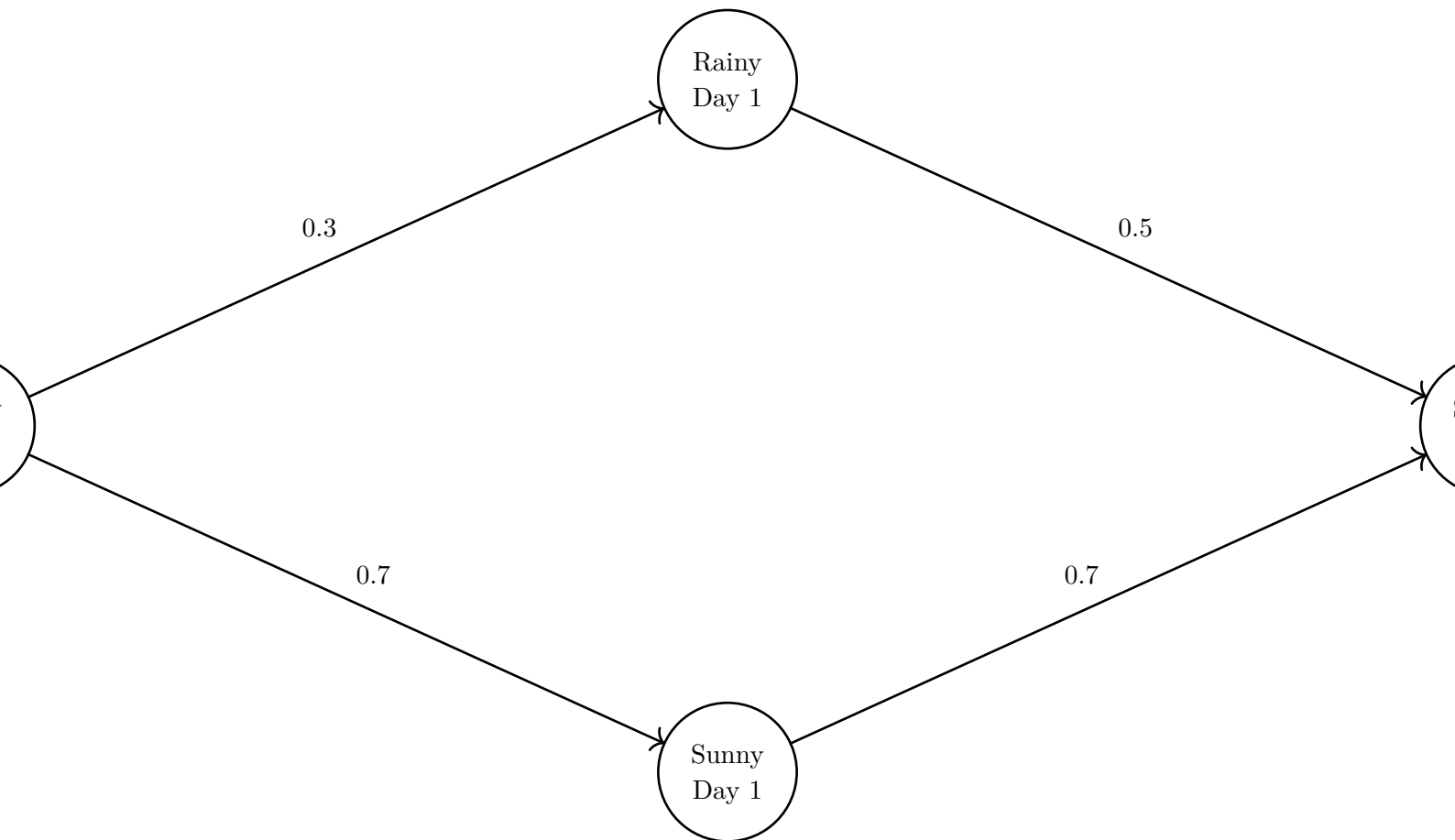
$$P^2 = \begin{pmatrix} 0.7 & 0.3 \\ 0.5 & 0.5 \end{pmatrix} \begin{pmatrix} 0.7 & 0.3 \\ 0.5 & 0.5 \end{pmatrix} = \begin{pmatrix} 0.64 & 0.36 \\ 0.60 & 0.40 \end{pmatrix}$$

The probability is $p_{SS}^{(2)} = 0.64 = 64\%$.

Detailed Calculation:

$$\begin{aligned} p_{SS}^{(2)} &= p_{SS} \cdot p_{SS} + p_{SR} \cdot p_{RS} \\ &= (0.7)(0.7) + (0.3)(0.5) = 0.49 + 0.15 = 0.64 \end{aligned}$$

Path Interpretation:



Path	States	Probability	Contribution
Path 1	$S \rightarrow S \rightarrow S$	$0.7 \times 0.7 = 0.49$	49%
Path 2	$S \rightarrow R \rightarrow S$	$0.3 \times 0.5 = 0.15$	15%
Total			64%

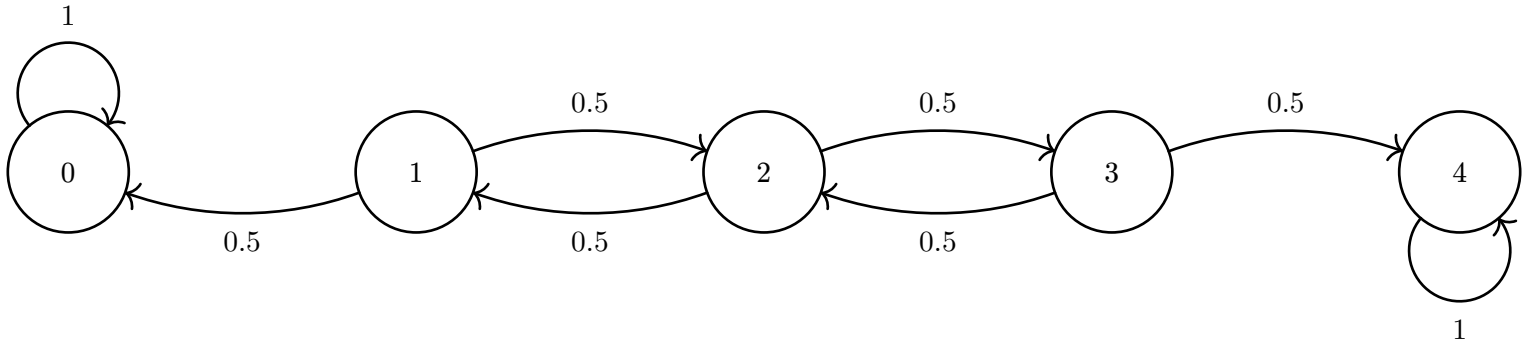
n-Step Transition Probabilities:

Steps (n)	P_{SS}^n	P_{SR}^n	P_{RS}^n	P_{RR}^n
0	1.000	0.000	0.000	1.000
1	0.700	0.300	0.500	0.500
2	0.640	0.360	0.600	0.400
3	0.628	0.372	0.620	0.380
5	0.6256	0.3744	0.6240	0.3760
10	0.62502	0.37498	0.62497	0.37503
∞	0.625	0.375	0.625	0.375

4.3 Example 2: Random Walk

A particle moves on the integers $\{0, 1, 2, 3, 4\}$ with:

- From state i (where $1 \leq i \leq 3$): move to $i + 1$ with probability 0.5, to $i - 1$ with probability 0.5
- States 0 and 4 are **absorbing states** (once reached, never leave)



Transition Matrix:

$$P = \begin{pmatrix} 1 & 0 & 0 & 0 & 0 \\ 0.5 & 0 & 0.5 & 0 & 0 \\ 0 & 0.5 & 0 & 0.5 & 0 \\ 0 & 0 & 0.5 & 0 & 0.5 \\ 0 & 0 & 0 & 0 & 1 \end{pmatrix}$$

4.4 Example 3: PageRank Algorithm

Google's PageRank algorithm uses Markov chains to rank web pages. Each webpage is a state, and transition probabilities are based on hyperlinks between pages.

Key idea:

- A page is important if it is linked to by other important pages
- Random surfer model: follow links randomly or jump to random page

The PageRank of page i is:

$$\text{PageRank}(i) = \frac{1-d}{N} + d \sum_{j \rightarrow i} \frac{\text{PageRank}(j)}{L(j)}$$

where:

- $d \approx 0.85$ is the damping factor
- N is total number of pages
- $L(j)$ is number of outgoing links from page j

5 Properties of States

5.1 Accessibility and Communication

Definition: State j is **accessible** from state i (written $i \rightarrow j$) if there exists $n \geq 0$ such that $p_{ij}^{(n)} > 0$.

Definition: States i and j **communicate** (written $i \leftrightarrow j$) if $i \rightarrow j$ and $j \rightarrow i$.

Communication is an equivalence relation:

- **Reflexive:** $i \leftrightarrow i$
- **Symmetric:** If $i \leftrightarrow j$, then $j \leftrightarrow i$
- **Transitive:** If $i \leftrightarrow j$ and $j \leftrightarrow k$, then $i \leftrightarrow k$

5.1.1 Communicating Classes: Detailed Analysis

The state space can be partitioned into **communicating classes**—maximal sets of states that communicate with each other.

Definition: A communicating class C is a set of states where:

1. All states in C communicate with each other
2. No state in C communicates with any state outside C

Finding Communicating Classes: Step-by-Step

Algorithm:

1. Start with any state i
2. Find all states accessible from i
3. Find all states from which i is accessible
4. The intersection forms a communicating class
5. Repeat for uncategorized states

Example 1: Identifying Classes

Consider the transition matrix:

$$P = \begin{pmatrix} 0.5 & 0.5 & 0 & 0 & 0 \\ 0.3 & 0.7 & 0 & 0 & 0 \\ 0 & 0 & 0.6 & 0.4 & 0 \\ 0 & 0 & 0.3 & 0.7 & 0 \\ 0.1 & 0 & 0 & 0 & 0.9 \end{pmatrix}$$

Step 1: Build accessibility table

From To	0	1	2	3	4
0	✓	✓	✗	✗	✗
1	✓	✓	✗	✗	✗
2	✗	✗	✓	✓	✗
3	✗	✗	✓	✓	✗
4	✓	✗	✗	✗	✓

Step 2: Identify classes

Class 1: $\{0, 1\}$

- $0 \leftrightarrow 1$ (can reach each other)
- Cannot reach states 2, 3, 4

Class 2: $\{2, 3\}$

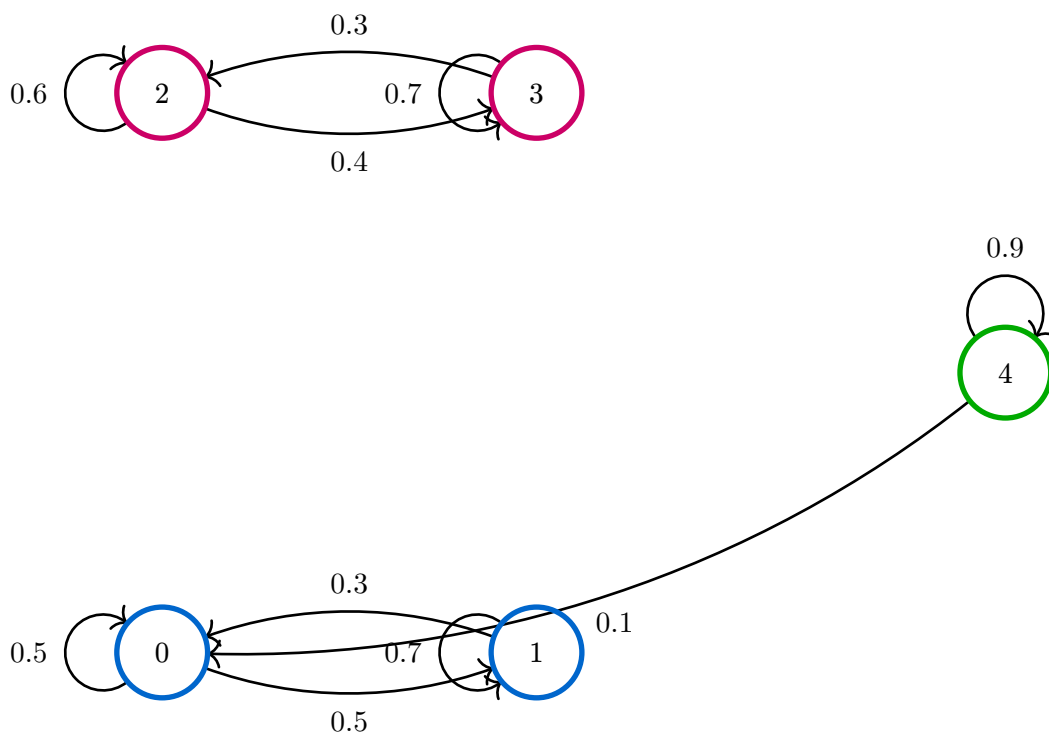
- $2 \leftrightarrow 3$ (can reach each other)
- Cannot reach other states

Class 3: $\{4\}$

- State 4 can reach state 0, but 0 cannot reach 4
- Forms its own class (transient)

Step 3: Classify classes

Class	States	Type	Reason
Class 1	$\{0, 1\}$	Closed	No exits possible
Class 2	$\{2, 3\}$	Closed	No exits possible
Class 3	$\{4\}$	Transient	Can exit to Class 1

State Diagram:

Example 2: Reducible vs Irreducible

Reducible Chain: Has multiple communicating classes

The chain above is reducible (3 classes).

Irreducible Chain: Only one communicating class (all states communicate)

$$P = \begin{pmatrix} 0 & 1 & 0 \\ 0 & 0 & 1 \\ 1 & 0 & 0 \end{pmatrix}$$

All states form one class: $\{0, 1, 2\}$

- $0 \rightarrow 1 \rightarrow 2 \rightarrow 0$ (cycle)
- All states can reach all others

Closed Classes and Absorption

Definition: A communicating class C is **closed** if:

$$p_{ij} = 0 \quad \text{for all } i \in C, j \notin C$$

Once the chain enters a closed class, it never leaves!

Theorem: Every finite Markov chain has at least one closed communicating class.

Long-term Behavior:

- Chain will eventually be absorbed into one of the closed classes
- Probability of absorption depends on initial state and transition probabilities

5.2 Irreducibility

A Markov chain is **irreducible** if all states communicate with each other. In other words, there is only one communicating class.

Example of Irreducible Chain:

$$P = \begin{pmatrix} 0 & 1 & 0 \\ 0.5 & 0 & 0.5 \\ 0 & 1 & 0 \end{pmatrix}$$

All states are mutually accessible.

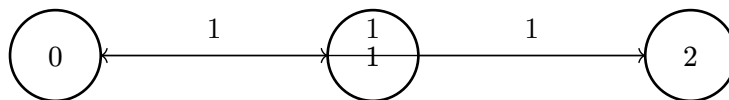
5.3 Periodicity

The **period** of state i is:

$$d(i) = \gcd\{n \geq 1 : p_{ii}^{(n)} > 0\}$$

- If $d(i) = 1$, state i is **aperiodic**
- If $d(i) > 1$, state i is **periodic** with period $d(i)$

Example of Periodic Chain:



This chain has period 3 (returns to state 0 only at steps 3, 6, 9, ...).

5.4 Recurrence and Transience

Definition: Let f_{ii} be the probability that the chain ever returns to state i , starting from i .

- State i is **recurrent** if $f_{ii} = 1$
- State i is **transient** if $f_{ii} < 1$

5.4.1 Classification of Recurrent States

Type	Definition
Positive Recurrent	Expected return time is finite: $E[T_i] < \infty$
Null Recurrent	Expected return time is infinite: $E[T_i] = \infty$

where T_i is the first return time to state i .

Key Result: For finite-state Markov chains, all recurrent states are positive recurrent.

5.5 Absorbing States

Definition: State i is **absorbing** if $p_{ii} = 1$ (once entered, never left).

An absorbing state is:

- Recurrent (obviously returns with probability 1)
- Aperiodic (period = 1)

6 Stationary Distributions

6.1 Definition

A probability distribution $\pi = [\pi_1, \pi_2, \dots, \pi_N]$ is a **stationary distribution** (or **steady-state distribution**) if:

$$\pi P = \pi$$

Equivalently, π is a left eigenvector of P with eigenvalue 1.

6.1.1 Properties

1. If $X_0 \sim \pi$, then $X_n \sim \pi$ for all $n \geq 0$
2. $\sum_{i=1}^N \pi_i = 1$
3. $\pi_i \geq 0$ for all i

6.2 Existence and Uniqueness

Theorem: An irreducible, aperiodic Markov chain with a finite state space has a unique stationary distribution π .

Furthermore:

$$\lim_{n \rightarrow \infty} P^n = \begin{pmatrix} \pi_1 & \pi_2 & \dots & \pi_N \\ \pi_1 & \pi_2 & \dots & \pi_N \\ \vdots & \vdots & \ddots & \vdots \\ \pi_1 & \pi_2 & \dots & \pi_N \end{pmatrix}$$

6.3 Computing Stationary Distributions

This section presents four comprehensive methods for finding stationary distributions, each with complete derivations and examples.

6.3.1 Method 1: Direct Linear System Solution

Mathematical Foundation:

The stationary distribution π satisfies:

1. Balance equation: $\pi P = \pi$
2. Normalization: $\sum_{i=1}^N \pi_i = 1$

Complete Algorithm:

Step 1: Write out the balance equations

$$\pi_j = \sum_{i=1}^N \pi_i p_{ij} \quad \text{for all } j = 1, 2, \dots, N$$

Step 2: Expand the system

$$\begin{cases} \pi_1 p_{11} + \pi_2 p_{21} + \dots + \pi_N p_{N1} = \pi_1 \\ \pi_1 p_{12} + \pi_2 p_{22} + \dots + \pi_N p_{N2} = \pi_2 \\ \vdots \\ \pi_1 p_{1N} + \pi_2 p_{2N} + \dots + \pi_N p_{NN} = \pi_N \end{cases}$$

Step 3: Rearrange to homogeneous form

$$\begin{cases} \pi_1(p_{11} - 1) + \pi_2 p_{21} + \dots + \pi_N p_{N1} = 0 \\ \pi_1 p_{12} + \pi_2(p_{22} - 1) + \dots + \pi_N p_{N2} = 0 \\ \vdots \\ \pi_1 p_{1N} + \pi_2 p_{2N} + \dots + \pi_N(p_{NN} - 1) = 0 \end{cases}$$

Step 4: Replace one equation with $\sum_i \pi_i = 1$

Step 5: Solve the resulting linear system

Detailed Example - 3-State Chain:

Consider:

$$P = \begin{pmatrix} 0.5 & 0.3 & 0.2 \\ 0.4 & 0.4 & 0.2 \\ 0.3 & 0.3 & 0.4 \end{pmatrix}$$

Step-by-step solution:

Balance equations:

$$\begin{cases} 0.5\pi_1 + 0.4\pi_2 + 0.3\pi_3 = \pi_1 \\ 0.3\pi_1 + 0.4\pi_2 + 0.3\pi_3 = \pi_2 \\ 0.2\pi_1 + 0.2\pi_2 + 0.4\pi_3 = \pi_3 \end{cases}$$

Rearranging:

$$\begin{cases} -0.5\pi_1 + 0.4\pi_2 + 0.3\pi_3 = 0 \\ 0.3\pi_1 - 0.6\pi_2 + 0.3\pi_3 = 0 \\ 0.2\pi_1 + 0.2\pi_2 - 0.6\pi_3 = 0 \end{cases}$$

Using equations 1 and 2 with normalization: From equation 1: $\pi_1 = \frac{0.4\pi_2 + 0.3\pi_3}{0.5} = 0.8\pi_2 + 0.6\pi_3$

From equation 2: $0.3\pi_1 = 0.6\pi_2 - 0.3\pi_3$ $\pi_1 = 2\pi_2 - \pi_3$

Equating: $0.8\pi_2 + 0.6\pi_3 = 2\pi_2 - \pi_3$ $1.6\pi_3 = 1.2\pi_2$ $\pi_3 = 0.75\pi_2$

Therefore: $\pi_1 = 2\pi_2 - 0.75\pi_2 = 1.25\pi_2$

Normalization: $\pi_1 + \pi_2 + \pi_3 = 1$ $1.25\pi_2 + \pi_2 + 0.75\pi_2 = 1$ $3\pi_2 = 1$ $\pi_2 = \frac{1}{3}$

Final solution: $\pi = [1.25 \cdot \frac{1}{3}, \frac{1}{3}, 0.75 \cdot \frac{1}{3}] = [\frac{5}{12}, \frac{1}{3}, \frac{1}{4}]$

State	π Value	Decimal	Percentage
1	$\frac{5}{12}$	0.4167	41.67%
2	$\frac{1}{3}$	0.3333	33.33%
3	$\frac{1}{4}$	0.2500	25.00%

6.3.2 Method 2: Eigenvalue/Eigenvector Approach

Theoretical Foundation:

The stationary distribution is the left eigenvector of P corresponding to eigenvalue $\lambda = 1$.

Complete Derivation:

Key Insight: $\pi P = \pi$ can be written as $\pi P = 1 \cdot \pi$

This means π is a left eigenvector of P with eigenvalue 1.

Equivalently: Solve $(P^T - I)\pi^T = 0$

Why eigenvalue 1 exists:

- P is stochastic: columns sum to 1
- $(P - I)e = 0$ where $e = [1, 1, \dots, 1]^T$
- So 1 is an eigenvalue of P (right eigenvector e)
- By Perron-Frobenius theorem, for irreducible P , the eigenvalue 1 has multiplicity 1

Step-by-step Algorithm:

1. Form matrix $A = P^T - I$
2. Solve $A\pi^T = 0$ (find null space)
3. Normalize: $\pi \leftarrow \frac{\pi}{\sum_i \pi_i}$

Detailed Example - Same 3-State Chain:

$$P = \begin{pmatrix} 0.5 & 0.3 & 0.2 \\ 0.4 & 0.4 & 0.2 \\ 0.3 & 0.3 & 0.4 \end{pmatrix}$$

Step 1: Compute P^T :

$$P^T = \begin{pmatrix} 0.5 & 0.4 & 0.3 \\ 0.3 & 0.4 & 0.3 \\ 0.2 & 0.2 & 0.4 \end{pmatrix}$$

Step 2: Form $A = P^T - I$:

$$A = \begin{pmatrix} -0.5 & 0.4 & 0.3 \\ 0.3 & -0.6 & 0.3 \\ 0.2 & 0.2 & -0.6 \end{pmatrix}$$

Step 3: Row reduce to find null space:

$$\begin{pmatrix} -0.5 & 0.4 & 0.3 \\ 0.3 & -0.6 & 0.3 \\ 0.2 & 0.2 & -0.6 \end{pmatrix} \rightarrow \begin{pmatrix} 1 & -0.8 & -0.6 \\ 0.3 & -0.6 & 0.3 \\ 0.2 & 0.2 & -0.6 \end{pmatrix}$$

After row operations:

$$\begin{pmatrix} 1 & -0.8 & -0.6 \\ 0 & -0.36 & 0.48 \\ 0 & 0.36 & -0.48 \end{pmatrix}$$

From row 2: $-0.36\pi_2 + 0.48\pi_3 = 0 \Rightarrow \pi_2 = \frac{0.48}{0.36}\pi_3 = \frac{4}{3}\pi_3$

From row 1: $\pi_1 = 0.8\pi_2 + 0.6\pi_3 = 0.8 \cdot \frac{4}{3}\pi_3 + 0.6\pi_3$ $\pi_1 = \frac{16}{15}\pi_3 + \frac{9}{15}\pi_3 = \frac{25}{15}\pi_3 = \frac{5}{3}\pi_3$

Step 4: Normalize with $\pi_1 + \pi_2 + \pi_3 = 1$: $\frac{5}{3}\pi_3 + \frac{4}{3}\pi_3 + \pi_3 = 1$ $\frac{5+4+3}{3}\pi_3 = 1$ $\pi_3 = \frac{3}{12} = \frac{1}{4}$

Therefore: $\pi = [\frac{5}{3} \cdot \frac{1}{4}, \frac{4}{3} \cdot \frac{1}{4}, \frac{1}{4}] = [\frac{5}{12}, \frac{1}{3}, \frac{1}{4}]$ ✓

Method	Advantage	Disadvantage
Direct Linear System	Intuitive, straightforward	May have numerical instability
Eigenvalue Approach	Theoretically elegant	Requires matrix computation

6.3.3 Method 3: Iterative Power Method

Algorithm Foundation:

For an irreducible, aperiodic chain: $\lim_{n \rightarrow \infty} P^n = \begin{pmatrix} \pi \\ \pi \\ \vdots \\ \pi \end{pmatrix}$

Each row converges to the stationary distribution.

Complete Algorithm:

Input: Transition matrix P , tolerance ε , initial distribution $\pi^{(0)}$

Iteration:

$$\pi^{(k+1)} = \pi^{(k)} P$$

Stopping criterion:

$$\|\pi^{(k+1)} - \pi^{(k)}\| < \varepsilon$$

Output: $\pi \approx \pi^{(k)}$

Convergence Rate:

For irreducible, aperiodic chain with eigenvalues $1 = \lambda_1 > |\lambda_2| \geq |\lambda_3| \geq \dots \geq |\lambda_N|$:

$$\|\pi^{(k)} - \pi\| \leq C |\lambda_2|^k$$

where C is a constant depending on the initial distribution.

Detailed Example with Convergence Table:

Using the same 3-state chain with $\pi^{(0)} = [1, 0, 0]$:

$$P = \begin{pmatrix} 0.5 & 0.3 & 0.2 \\ 0.4 & 0.4 & 0.2 \\ 0.3 & 0.3 & 0.4 \end{pmatrix}$$

Iteration calculations:

$$k = 1: \pi^{(1)} = [1, 0, 0] \begin{pmatrix} 0.5 & 0.3 & 0.2 \\ 0.4 & 0.4 & 0.2 \\ 0.3 & 0.3 & 0.4 \end{pmatrix} = [0.5, 0.3, 0.2]$$

$$k = 2: \pi^{(2)} = [0.5, 0.3, 0.2] P = [0.43, 0.33, 0.24]$$

$$k = 3: \pi^{(3)} = [0.43, 0.33, 0.24] P = [0.424, 0.330, 0.246]$$

$$k = 4: \pi^{(4)} = [0.424, 0.330, 0.246] P = [0.4196, 0.3326, 0.2478]$$

Iteration k	$\pi_1^{(k)}$	$\pi_2^{(k)}$	$\pi_3^{(k)}$	Max Error
0	1.0000	0.0000	0.0000	0.5833
1	0.5000	0.3000	0.2000	0.0833
2	0.4300	0.3300	0.2400	0.0133
3	0.4240	0.3300	0.2460	0.0073
4	0.4196	0.3326	0.2478	0.0029
5	0.4177	0.3329	0.2494	0.0010
10	0.4168	0.3333	0.2499	0.0001
∞	0.4167	0.3333	0.2500	0.0000
Exact	$\frac{5}{12}$	$\frac{1}{3}$	$\frac{1}{4}$	-

Convergence Analysis:

- Geometric rate of convergence
- Error decreases by factor of approximately $|\lambda_2| \approx 0.2$ per iteration
- Practical convergence (error < 0.001) achieved in 10 iterations

6.3.4 Method 4: Detailed Balance Equations (for Reversible Chains)**Definition of Reversibility:**

A Markov chain is **reversible** with respect to distribution π if:

$$\pi_i p_{ij} = \pi_j p_{ji} \quad \text{for all states } i, j$$

This is called the **detailed balance condition**.

Physical Interpretation: The probability flow from i to j equals the flow from j to i in equilibrium.

Why This Simplifies Computation:

If detailed balance holds, we can solve $\frac{N(N-1)}{2}$ equations instead of N equations!

Complete Algorithm for Reversible Chains:

1. Verify reversibility by checking if $\frac{p_{ij}}{p_{ji}}$ is consistent across paths
2. Choose a reference state (usually state 1)
3. For each state j , compute:

$$\pi_j = \pi_1 \cdot \prod_{\text{path } 1 \rightarrow j} \frac{p_k}{p_{k-1}}$$

4. Normalize: $\pi_1 = \frac{1}{1 + \sum_{j \neq 1} \frac{\pi_j}{\pi_1}}$

Detailed Example - Birth-Death Process:

A birth-death process has:

- Birth rates: $\lambda_0 = 2, \lambda_1 = 1.5, \lambda_2 = 1$
- Death rates: $\mu_1 = 1, \mu_2 = 1.5, \mu_3 = 2$

Transition probabilities (normalizing within each row):

$$P = \begin{pmatrix} 0 & \frac{2}{3} & \frac{1}{3} & 0 \\ \frac{1}{2.5} & 0 & \frac{1.5}{2.5} & 0 \\ 0 & \frac{1.5}{2.5} & 0 & \frac{1}{2.5} \\ 0 & 0 & \frac{2}{3} & \frac{1}{3} \end{pmatrix}$$

Wait, let me construct this more carefully:

From state 0: can go to 1 with rate $\lambda_0 = 2$ From state 1: can go to 0 with rate $\mu_1 = 1$, or to 2 with rate $\lambda_1 = 1.5$ Total rate from 1: $\mu_1 + \lambda_1 = 2.5$

Detailed balance approach:

For birth-death, detailed balance gives:

$$\pi_i \lambda_i = \pi_{i+1} \mu_{i+1}$$

Therefore:

$$\pi_1 = \pi_0 \cdot \frac{\lambda_0}{\mu_1} = \pi_0 \cdot \frac{2}{1} = 2\pi_0$$

$$\pi_2 = \pi_1 \cdot \frac{\lambda_1}{\mu_2} = 2\pi_0 \cdot \frac{1.5}{1.5} = 2\pi_0$$

$$\pi_3 = \pi_2 \cdot \frac{\lambda_2}{\mu_3} = 2\pi_0 \cdot \frac{1}{2} = \pi_0$$

Normalization:

$$\pi_0 + \pi_1 + \pi_2 + \pi_3 = 1$$

$$\pi_0 + 2\pi_0 + 2\pi_0 + \pi_0 = 1$$

$$6\pi_0 = 1$$

$$\pi_0 = \frac{1}{6}$$

Final stationary distribution:

$$\pi = \left[\frac{1}{6}, \frac{2}{6}, \frac{2}{6}, \frac{1}{6} \right] = \left[\frac{1}{6}, \frac{1}{3}, \frac{1}{3}, \frac{1}{6} \right]$$

State	π Value	Detailed Balance Verification	Flow Rate
0	1/6	$\pi_0 \lambda_0 = \frac{1}{6} \cdot 2 = \frac{1}{3}$	$\rightarrow 1: 1/3$
1	1/3	$\pi_1 \mu_1 = \frac{1}{3} \cdot 1 = \frac{1}{3} \checkmark$	$\rightarrow 0: 1/3$
		$\pi_1 \lambda_1 = \frac{1}{3} \cdot 1.5 = \frac{1}{2}$	$\rightarrow 2: 1/2$
2	1/3	$\pi_2 \mu_2 = \frac{1}{3} \cdot 1.5 = \frac{1}{2} \checkmark$	$\rightarrow 1: 1/2$
		$\pi_2 \lambda_2 = \frac{1}{3} \cdot 1 = \frac{1}{3}$	$\rightarrow 3: 1/3$
3	1/6	$\pi_3 \mu_3 = \frac{1}{6} \cdot 2 = \frac{1}{3} \checkmark$	$\rightarrow 2: 1/3$

Summary of Four Methods:

Method	Best For	Complexity	Special Requirements
--------	----------	------------	----------------------

Direct Linear System	Small chains ($N \leq 10$)	$O(N^3)$	None
Eigenvalue Approach	Theoretical analysis	$O(N^3)$	None
Power Iteration	Large sparse chains	$O(N^2k)$	Aperiodic chain
Detailed Balance	Reversible chains	$O(N)$	Reversibility

Practical Recommendations:

1. For small chains: Use direct method or eigenvalue approach
2. For large sparse chains: Use power iteration
3. For birth-death or random walks: Check detailed balance first
4. Always verify solution by checking $\pi P = \pi$ and $\sum_i \pi_i = 1$

6.3.5 Example: Weather Model (Detailed Solution)

For our weather example:

$$P = \begin{pmatrix} 0.7 & 0.3 \\ 0.5 & 0.5 \end{pmatrix}$$

Solve $\pi P = \pi$:

$$[\pi_S, \pi_R] \begin{pmatrix} 0.7 & 0.3 \\ 0.5 & 0.5 \end{pmatrix} = [\pi_S, \pi_R]$$

Step-by-step derivation:

Expanding the matrix equation:

$$\begin{cases} \pi_S \cdot 0.7 + \pi_R \cdot 0.5 = \pi_S \\ \pi_S \cdot 0.3 + \pi_R \cdot 0.5 = \pi_R \\ \pi_S + \pi_R = 1 \end{cases}$$

From equation 1:

$$0.7\pi_S + 0.5\pi_R = \pi_S$$

$$0.5\pi_R = \pi_S - 0.7\pi_S$$

$$0.5\pi_R = 0.3\pi_S$$

$$\pi_R = \frac{0.3}{0.5}\pi_S = \frac{3}{5}\pi_S$$

Substituting into $\pi_S + \pi_R = 1$:

$$\pi_S + \frac{3}{5}\pi_S = 1$$

$$\frac{5}{5}\pi_S + \frac{3}{5}\pi_S = 1$$

$$\frac{8}{5}\pi_S = 1$$

$$\pi_S = \frac{5}{8} = 0.625$$

Therefore:

$$\pi_R = 1 - \pi_S = 1 - \frac{5}{8} = \frac{3}{8} = 0.375$$

Verification:

$$\begin{aligned} \left[\frac{5}{8}, \frac{3}{8} \right] \begin{pmatrix} 0.7 & 0.3 \\ 0.5 & 0.5 \end{pmatrix} &= \left[\frac{5}{8} \cdot 0.7 + \frac{3}{8} \cdot 0.5, \frac{5}{8} \cdot 0.3 + \frac{3}{8} \cdot 0.5 \right] \\ &= [0.4375 + 0.1875, 0.1875 + 0.1875] = [0.625, 0.375] = \left[\frac{5}{8}, \frac{3}{8} \right] \checkmark \end{aligned}$$

Measure	Sunny	Rainy
---------	-------	-------

Stationary Prob.	0.625 (62.5%)	0.375 (37.5%)
Expected Return Time	$1/0.625 = 1.6$ days	$1/0.375 \approx 2.67$ days
Interpretation	Sunny 5 out of 8 days	Rainy 3 out of 8 days

Interpretation: In the long run, about 62.5% of days are sunny and 37.5% are rainy.

7 Ergodic Markov Chains

7.1 Definition

A state is **ergodic** if it is:

1. Aperiodic
2. Positive recurrent

A Markov chain is **ergodic** if all its states are ergodic.

7.2 Ergodic Theorem

Theorem (Ergodic Theorem): For an irreducible, aperiodic Markov chain with stationary distribution π :

$$\lim_{n \rightarrow \infty} p_{ij}^{(n)} = \pi_j \quad \text{for all states } i, j$$

Furthermore:

$$\lim_{N \rightarrow \infty} \frac{1}{N} \sum_{n=1}^N \mathbb{1}\{X_n = j\} = \pi_j \quad \text{with probability 1}$$

This means:

- The proportion of time spent in state j converges to π_j
- Long-run average equals the stationary probability

7.2.1 Complete Proof of Ergodic Theorem

We prove this fundamental result in several steps, building from basic principles to the full convergence guarantee.

Proof Strategy:

The proof relies on three key ideas:

1. Spectral decomposition of the transition matrix
2. Properties of aperiodic, irreducible chains
3. Perron-Frobenius theorem for stochastic matrices

Step 1: Eigenvalue Structure

Lemma 1: For an irreducible, aperiodic, finite-state Markov chain with transition matrix P :

1. $\lambda = 1$ is an eigenvalue with multiplicity 1
2. All other eigenvalues satisfy $|\lambda_i| < 1$
3. The eigenvector corresponding to $\lambda = 1$ is the stationary distribution π

Proof sketch:

- Irreducibility + aperiodicity $\implies$ primitive matrix (by Perron-Frobenius)
- Primitive stochastic matrix has unique largest eigenvalue 1
- Left eigenvector for $\lambda = 1$ is stationary distribution: $\pi P = \pi$

Step 2: Spectral Decomposition

For finite-state chains, we can decompose:

$$P = \sum_{i=1}^N \lambda_i v_i w_i^T$$

where λ_i are eigenvalues, v_i right eigenvectors, w_i left eigenvectors.

For our ergodic chain:

$$P = 1 \cdot e\pi^T + \sum_{i=2}^N \lambda_i v_i w_i^T$$

where $e = [1, 1, \dots, 1]^T$ and π is the stationary distribution.

Taking n -th power:

$$P^n = 1^n \cdot e\pi^T + \sum_{i=2}^N \lambda_i^n v_i w_i^T = e\pi^T + \sum_{i=2}^N \lambda_i^n v_i w_i^T$$

Step 3: Convergence to Stationary Distribution

As $n \rightarrow \infty$:

$$\lim_{n \rightarrow \infty} P^n = \lim_{n \rightarrow \infty} \left[e\pi^T + \sum_{i=2}^N \lambda_i^n v_i w_i^T \right]$$

Since $|\lambda_i| < 1$ for $i \geq 2$ (by Lemma 1):

$$\lim_{n \rightarrow \infty} \lambda_i^n = 0 \quad \text{for all } i \geq 2$$

Therefore:

$$\lim_{n \rightarrow \infty} P^n = e\pi^T = \begin{pmatrix} \pi_1 & \pi_2 & \dots & \pi_N \\ \pi_1 & \pi_2 & \dots & \pi_N \\ \vdots & \vdots & \ddots & \vdots \\ \pi_1 & \pi_2 & \dots & \pi_N \end{pmatrix}$$

This proves:

$$\lim_{n \rightarrow \infty} p_{ij}^{(n)} = \pi_j \quad \text{for all } i, j$$

Step 4: Convergence Rate

The rate of convergence is determined by the second-largest eigenvalue magnitude:

$$\rho = \max\{|\lambda_2|, |\lambda_3|, \dots, |\lambda_N|\}$$

We can bound the error:

$$|p_{ij}^{(n)} - \pi_j| \leq C\rho^n$$

where C depends on the eigenvector structure.

Step 5: Time-Average Convergence (SLLN Part)

Theorem (Strong Law for Markov Chains):

$$\lim_{N \rightarrow \infty} \frac{1}{N} \sum_{n=1}^N \mathbb{1}\{X_n = j\} = \pi_j \quad \text{almost surely}$$

Proof outline:

Define $V_{N(j)} = \sum_{n=1}^N \mathbb{1}\{X_n = j\}$ (number of visits to state j by time N)

Claim: $\frac{V_{N(j)}}{N} \rightarrow \pi_j$ almost surely

1. By Markov property:

$$\mathbb{E}[V_{N(j)} \mid X_0 = i] = \sum_{n=1}^N p_{ij}^{(n)}$$

2. From Step 3, we know $p_{ij}^{(n)} \rightarrow \pi_j$, so:

$$\lim_{N \rightarrow \infty} \frac{\mathbb{E}[V_{N(j)}]}{N} = \lim_{N \rightarrow \infty} \frac{1}{N} \sum_{n=1}^N p_{ij}^{(n)} = \pi_j$$

3. By ergodic theorem for Markov chains (more general than SLLN):

$$\lim_{N \rightarrow \infty} \frac{V_{N(j)}}{N} = \pi_j \quad \text{almost surely}$$

This uses the fact that for ergodic chains, the empirical distribution converges to the stationary distribution.

Detailed Example: 2-State Weather Model

Let's verify convergence numerically:

$$P = \begin{pmatrix} 0.7 & 0.3 \\ 0.5 & 0.5 \end{pmatrix}$$

Eigenvalue analysis:

Characteristic polynomial: $\det(P - \lambda I) = 0$

$$\det \begin{pmatrix} 0.7 - \lambda & 0.3 \\ 0.5 & 0.5 - \lambda \end{pmatrix} = (0.7 - \lambda)(0.5 - \lambda) - 0.15 = 0$$

$$\lambda^2 - 1.2\lambda + 0.35 - 0.15 = 0$$

$$\lambda^2 - 1.2\lambda + 0.2 = 0$$

Using quadratic formula:

$$\lambda = \frac{1.2 \pm \sqrt{1.44 - 0.8}}{2} = \frac{1.2 \pm \sqrt{0.64}}{2} = \frac{1.2 \pm 0.8}{2}$$

So: $\lambda_1 = 1$, $\lambda_2 = 0.2$

Convergence rate: $\rho = |\lambda_2| = 0.2$

Numerical verification:

n	$p_{SS}^{(n)}$	$p_{SR}^{(n)}$	$p_{RS}^{(n)}$	$p_{RR}^{(n)}$	Max Error
---	----------------	----------------	----------------	----------------	-----------

0	1.000	0.000	0.000	1.000	0.3750
1	0.700	0.300	0.500	0.500	0.0750
2	0.640	0.360	0.600	0.400	0.0150
3	0.628	0.372	0.620	0.380	0.0030
5	0.6256	0.3744	0.6240	0.3760	0.0006
10	0.62500	0.37500	0.62500	0.37500	0.00001
∞	0.625	0.375	0.625	0.375	0.000
Exact π	$\frac{5}{8}$	$\frac{3}{8}$	$\frac{5}{8}$	$\frac{3}{8}$	-

Error decay verification:

For $n = 5$: Error $\approx 0.0006 \approx C \cdot 0.2^5 = C \cdot 0.00032$

The error decreases geometrically with rate $\rho = 0.2$, confirming the theorem.

Physical interpretation:

1. After 10 steps, any initial state “forgets” where it started
2. Distribution converges to $[0.625, 0.375]$ regardless of X_0
3. Over long runs, spend 62.5% of time sunny, 37.5% rainy
4. Convergence is exponentially fast with rate 0.2^n

Key insight: Aperiodicity ensures smooth convergence (no oscillations), irreducibility ensures all states eventually communicate, leading to unique stationary distribution.

7.3 Convergence Rate

The convergence to stationarity is geometric:

$$\|P^n - \pi\|_{\text{TV}} \leq C\rho^n$$

where:

- $\|\cdot\|_{\text{TV}}$ is total variation distance
- $\rho < 1$ is the second-largest eigenvalue of P
- C is a constant

Faster convergence occurs when ρ is small (eigenvalues well-separated).

8 Absorbing Markov Chains

8.1 Definition

A Markov chain is **absorbing** if:

1. It has at least one absorbing state
2. From every state, it is possible to reach an absorbing state

8.2 Canonical Form

An absorbing chain can be written in canonical form:

$$P = \begin{pmatrix} Q & R \\ 0 & I \end{pmatrix}$$

where:

- Q : transition probabilities between transient states
- R : transition probabilities from transient to absorbing states
- I : identity matrix (absorbing states stay put)
- 0 : zero matrix (can't leave absorbing states)

8.2.1 How to Construct Canonical Form: Complete Methodology

Step-by-Step Algorithm:

Step 1: Identify State Types

- Absorbing states: $p_{ii} = 1$
- Transient states: all others (can eventually reach absorbing states)

Step 2: Reorder States

- List all transient states first: $\{1, 2, \dots, t\}$
- Then list absorbing states: $\{t+1, t+2, \dots, t+a\}$
- Total states: $N = t + a$

Step 3: Extract Submatrices

- Q : $(t \times t)$ matrix of transitions between transient states
- R : $(t \times a)$ matrix from transient to absorbing states
- Verify: Each row of $[Q \mid R]$ sums to 1
- Bottom blocks: 0 is $(a \times t)$ zero matrix, I is $(a \times a)$ identity

Step 4: Verify Validity

- Check: Q has spectral radius < 1 (eigenvalues inside unit circle)
- Check: $(I - Q)$ is invertible
- Check: All entries of $(I - Q)^{-1}$ are non-negative

Detailed Example 1: Simple 4-State Absorbing Chain

Original transition matrix (states ordered 1, 2, 3, 4):

$$P_{\text{original}} = \begin{pmatrix} 0.5 & 0.3 & 0.2 & 0 \\ 0.4 & 0 & 0.6 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{pmatrix}$$

Step 1: Identify states

- Transient: states 1, 2 (can leave)
- Absorbing: states 3, 4 ($p_{33} = p_{44} = 1$)

Step 2: Already in canonical form! (transient first, absorbing last)

Step 3: Extract blocks

$$Q = \begin{pmatrix} 0.5 & 0.3 \\ 0.4 & 0 \end{pmatrix} \quad (\text{transitions among transient states 1,2})$$

$$R = \begin{pmatrix} 0.2 & 0 \\ 0.6 & 0 \end{pmatrix} \quad (\text{from transient to absorbing 3,4})$$

Verification:

- Row 1 of $[Q \mid R]$: $0.5 + 0.3 + 0.2 + 0 = 1 \checkmark$
- Row 2 of $[Q \mid R]$: $0.4 + 0 + 0.6 + 0 = 1 \checkmark$

Detailed Example 2: Reordering Required

Original matrix (states A, B, C, D where B and D are absorbing):

	A	B	C	D
A	0.3	0.4	0.2	0.1
B	0	1	0	0
C	0.5	0.2	0.1	0.2
D	0	0	0	1

$$P_{\text{original}} = \begin{pmatrix} 0.3 & 0.4 & 0.2 & 0.1 \\ 0 & 1 & 0 & 0 \\ 0.5 & 0.2 & 0.1 & 0.2 \\ 0 & 0 & 0 & 1 \end{pmatrix}$$

Step 1: Identify

- Transient: A (state 1), C (state 3)
- Absorbing: B (state 2), D (state 4)

Step 2: Reorder as (A, C, B, D)

New state order: $1 \rightarrow 1, 2 \rightarrow 3, 3 \rightarrow 2, 4 \rightarrow 4$

Permutation matrix construction:

To move from old to new order, apply permutation (1, 3, 2, 4):

$$P_{\text{canonical}} = \Pi P_{\text{original}} \Pi^T$$

where Π swaps rows/columns 2 and 3.

Step 3: Result in canonical form

	A	C	B	D	
A	0.3	0.2	0.4	0.1	
C	0.5	0.1	0.2	0.2	

—	—	—	—	—	
B	0	0	1	0	
D	0	0	0	1	

$$P_{\text{canonical}} = \left(\begin{array}{cc|cc} 0.3 & 0.2 & 0.4 & 0.1 \\ 0.5 & 0.1 & 0.2 & 0.2 \\ \hline 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{array} \right) = \begin{pmatrix} Q & R \\ 0 & I \end{pmatrix}$$

where:

$$Q = \begin{pmatrix} 0.3 & 0.2 \\ 0.5 & 0.1 \end{pmatrix} \quad R = \begin{pmatrix} 0.4 & 0.1 \\ 0.2 & 0.2 \end{pmatrix}$$

Verification of row sums:

- Row A: $0.3 + 0.2 + 0.4 + 0.1 = 1 \checkmark$
- Row C: $0.5 + 0.1 + 0.2 + 0.2 = 1 \checkmark$

Mathematical Properties of Canonical Form:

Theorem (Powers of Canonical Form):

$$P^n = \begin{pmatrix} Q^n & \left(\sum_{k=0}^{n-1} Q^k \right) R \\ 0 & I \end{pmatrix}$$

Proof by induction:

Base case (n=1): True by definition

Inductive step: Assume true for n , prove for $n + 1$:

$$P^{n+1} = P^n \cdot P = \begin{pmatrix} Q^n & \left(\sum_{k=0}^{n-1} Q^k \right) R \\ 0 & I \end{pmatrix} \begin{pmatrix} Q & R \\ 0 & I \end{pmatrix}$$

Block multiplication:

- Top-left: $Q^n \cdot Q = Q^{n+1} \checkmark$
- Top-right: $Q^n R + \left(\sum_{k=0}^{n-1} Q^k \right) R \cdot I = Q^n R + \sum_{k=0}^{n-1} Q^k R = \sum_{k=0}^n Q^k R \checkmark$
- Bottom-left: $0 \cdot Q + 0 = 0 \checkmark$
- Bottom-right: $0 \cdot R + I \cdot I = I \checkmark$

Convergence: As $n \rightarrow \infty$, $Q^n \rightarrow 0$ (since spectral radius < 1):

$$\lim_{n \rightarrow \infty} P^n = \begin{pmatrix} 0 & (I - Q)^{-1} R \\ 0 & I \end{pmatrix}$$

The top-right block $(I - Q)^{-1} R$ gives absorption probabilities!

Step-by-Step Calculation for Example 1:

$$Q = \begin{pmatrix} 0.5 & 0.3 \\ 0.4 & 0 \end{pmatrix}$$

Compute $I - Q$:

$$I - Q = \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix} - \begin{pmatrix} 0.5 & 0.3 \\ 0.4 & 0 \end{pmatrix} = \begin{pmatrix} 0.5 & -0.3 \\ -0.4 & 1 \end{pmatrix}$$

Compute $(I - Q)^{-1}$ using matrix inversion:

For $\begin{pmatrix} a & b \\ c & d \end{pmatrix}$, inverse is $\frac{1}{ad-bc} \begin{pmatrix} d & -b \\ -c & a \end{pmatrix}$

Determinant: $\det(I - Q) = 0.5 \cdot 1 - (-0.3)(-0.4) = 0.5 - 0.12 = 0.38$

$$(I - Q)^{-1} = \frac{1}{0.38} \begin{pmatrix} 1 & 0.3 \\ 0.4 & 0.5 \end{pmatrix} = \begin{pmatrix} 2.632 & 0.789 \\ 1.053 & 1.316 \end{pmatrix}$$

This is the fundamental matrix N !

8.3 Fundamental Matrix

The fundamental matrix is:

$$N = (I - Q)^{-1} = I + Q + Q^2 + Q^3 + \dots$$

Interpretation: N_{ij} is the expected number of times the chain visits transient state j starting from transient state i .

8.3.1 Absorption Probabilities

The matrix $B = NR$ gives absorption probabilities:

- B_{ij} = probability of being absorbed in state j starting from transient state i

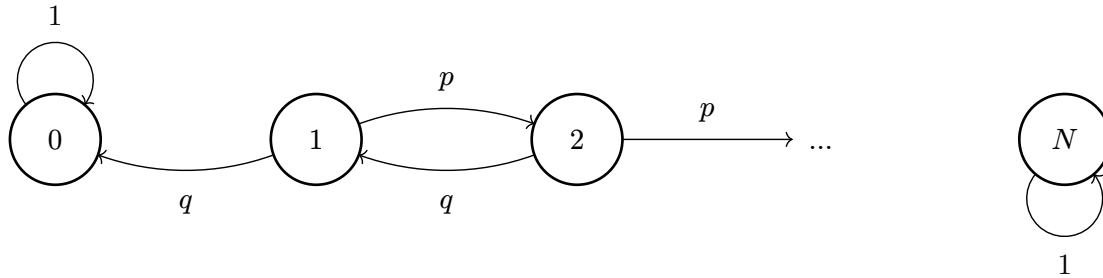
8.3.2 Expected Time to Absorption

The vector $t = N\mathbf{1}$ gives expected absorption times:

- t_i = expected number of steps until absorption, starting from state i

8.4 Example: Gambler's Ruin

A gambler starts with k dollars and plays until reaching 0 (broke) or N (wins). Each game, win \$1 with probability p , lose \$1 with probability $q = 1 - p$.



Questions of Interest:

1. Probability of ruin starting with k dollars?
2. Expected duration of the game?

8.4.1 Detailed Solution for Fair Game ($p = q = \frac{1}{2}$)

Part 1: Probability of Ruin

Let P_k = probability of ruin (reaching \$0) starting from k dollars.

Boundary conditions:

- $P_0 = 1$ (already ruined)
- $P_N = 0$ (already won)

Recursive relation: From state k ($1 \leq k \leq N - 1$):

$$P_k = p \cdot P_{k+1} + q \cdot P_{k-1}$$

For fair game ($p = q = \frac{1}{2}$):

$$P_k = \frac{1}{2}P_{k+1} + \frac{1}{2}P_{k-1}$$

$$2P_k = P_{k+1} + P_{k-1}$$

$$P_{k+1} - P_k = P_k - P_{k-1}$$

This means the differences are constant! Let $d = P_1 - P_0 = P_1 - 1$.

Then:

- $P_1 = P_0 + d = 1 + d$
- $P_2 = P_1 + d = 1 + 2d$
- $P_3 = P_2 + d = 1 + 3d$
- $P_k = 1 + kd$

Using $P_N = 0$:

$$0 = 1 + Nd$$

$$d = -\frac{1}{N}$$

Therefore:

$$P_k = 1 + k \left(-\frac{1}{N} \right) = 1 - \frac{k}{N} = \frac{N-k}{N}$$

Numerical Example: If $N = 10$ and $k = 3$:

$$P_3 = \frac{10-3}{10} = \frac{7}{10} = 0.7$$

70% chance of ruin, 30% chance of winning.

Starting Amount (k)	P(Ruin)	P(Win)	Odds
1	$9/10 = 0.90$	0.10	9:1 against
3	$7/10 = 0.70$	0.30	7:3 against
5	$5/10 = 0.50$	0.50	Even
7	$3/10 = 0.30$	0.70	3:7 for
9	$1/10 = 0.10$	0.90	1:9 for

Part 2: Expected Duration

Let D_k = expected number of games starting from k dollars.

Boundary conditions:

- $D_0 = 0$ (game over)
- $D_N = 0$ (game over)

Recursive relation:

$$D_k = 1 + p \cdot D_{k+1} + q \cdot D_{k-1}$$

The “+1” accounts for the current game.

For fair game ($p = q = \frac{1}{2}$):

$$D_k = 1 + \frac{1}{2}D_{k+1} + \frac{1}{2}D_{k-1}$$

$$2D_k = 2 + D_{k+1} + D_{k-1}$$

$$D_{k+1} - 2D_k + D_{k-1} = -2$$

This is a second-order difference equation. Try solution $D_k = ak^2 + bk + c$:

$$D_{k+1} = a(k+1)^2 + b(k+1) + c = ak^2 + 2ak + a + bk + b + c$$

$$D_{k-1} = a(k-1)^2 + b(k-1) + c = ak^2 - 2ak + a + bk - b + c$$

Substituting:

$$(ak^2 + 2ak + a + bk + b + c) - 2(ak^2 + bk + c) + (ak^2 - 2ak + a + bk - b + c) = -2$$

$$2a = -2$$

$$a = -1$$

So $D_k = -k^2 + bk + c$.

Using $D_0 = 0$: $c = 0$ Using $D_N = 0$: $-N^2 + bN = 0$, so $b = N$

Therefore:

$$D_k = -k^2 + Nk = k(N - k)$$

Numerical Example: If $N = 10$ and $k = 3$:

$$D_3 = 3(10 - 3) = 3 \cdot 7 = 21 \text{ games}$$

Starting Amount (k)	Expected Duration	Formula
1	$1(10-1) = 9$	$k(N - k)$
2	$2(10-2) = 16$	
3	$3(10-3) = 21$	
5	$5(10-5) = 25$	Maximum at $k = \frac{N}{2}$
7	$7(10-7) = 21$	
9	$9(10-9) = 9$	

Key Insight: Expected duration is maximized when starting at the midpoint ($k = \frac{N}{2}$).

8.4.2 General Case (Unfair Game, $p \neq q$)

For $p \neq \frac{1}{2}$, let $r = \frac{q}{p}$:

Probability of ruin:

$$P_k = \begin{cases} \frac{r^N - r^k}{r^N - 1} & \text{if } p \neq \frac{1}{2} \\ \frac{N-k}{N} & \text{if } p = \frac{1}{2} \end{cases}$$

Expected duration:

$$D_k = \begin{cases} \frac{1}{q-p} \left(k - N \frac{r^k - 1}{r^N - 1} \right) & \text{if } p \neq \frac{1}{2} \\ k(N - k) & \text{if } p = \frac{1}{2} \end{cases}$$

Example with advantage ($p = 0.6, q = 0.4$):

For $N = 10, k = 5, r = \frac{0.4}{0.6} = \frac{2}{3}$:

$$P_5 = \frac{\left(\frac{2}{3}\right)^{10} - \left(\frac{2}{3}\right)^5}{\left(\frac{2}{3}\right)^{10} - 1} \approx 0.087$$

Much better than fair game! Only 8.7% chance of ruin vs 50%.

Game Type	p	P(Ruin) from \$5	Interpretation
Player disadvantage	0.4	0.913	91.3% lose
Fair game	0.5	0.500	50% lose
Player advantage	0.6	0.087	8.7% lose

9 First Passage Times and Hitting Probabilities

This section explores the fundamental questions: “How long until we reach a target state?” and “What’s the probability we ever reach it?”

9.1 Definitions

First Passage Time from state i to state j :

$$T_j^{(i)} = \min\{n \geq 1 : X_n = j \mid X_0 = i\}$$

First Return Time to state i (when $i = j$):

$$T_i = \min\{n \geq 1 : X_n = i \mid X_0 = i\}$$

Hitting Probability (or reachability):

$$h_{ij} = P(T_j^{(i)} < \infty) = P(\text{ever reach } j \text{ from } i)$$

Mean First Passage Time:

$$m_{ij} = \mathbb{E}[T_j^{(i)} \mid T_j^{(i)} < \infty]$$

Mean Return Time to state i :

$$\mu_i = \mathbb{E}[T_i \mid X_0 = i] = m_{ii}$$

9.2 Computing Hitting Probabilities

Method 1: First-Step Analysis

The hitting probability h_{ij} satisfies:

Boundary conditions:

$$h_{jj} = 1 \quad (\text{already at target})$$

Recursive equation: For $i \neq j$:

$$h_{ij} = \sum_{k \in S} p_{ik} h_{kj}$$

System form: This gives a linear system with $|S| - 1$ equations (excluding j).

Matrix formulation: Let $h = [h_{1j}, h_{2j}, \dots, h_{Nj}]^T$

Then: $(I - \tilde{P})h = b$ where $\tilde{P}$ is P with row/column j removed, b has 1's for states that can reach j directly.

Detailed Example 1: 3-State Chain

Consider:

$$P = \begin{pmatrix} 0.5 & 0.3 & 0.2 \\ 0.4 & 0.4 & 0.2 \\ 0 & 0 & 1 \end{pmatrix}$$

State 3 is absorbing. Find h_{13} and h_{23} .

Solution using first-step analysis:

Since state 3 is absorbing: $h_{33} = 1$

For state 1:

$$h_{13} = 0.5 \cdot h_{13} + 0.3 \cdot h_{23} + 0.2 \cdot h_{33}$$

$$h_{13} = 0.5h_{13} + 0.3h_{23} + 0.2$$

For state 2:

$$h_{23} = 0.4 \cdot h_{13} + 0.4 \cdot h_{23} + 0.2 \cdot h_{33}$$

$$h_{23} = 0.4h_{13} + 0.4h_{23} + 0.2$$

Rearranging into standard form:

$$\begin{cases} 0.5h_{13} - 0.3h_{23} = 0.2 \\ -0.4h_{13} + 0.6h_{23} = 0.2 \end{cases}$$

From equation 1: $h_{13} = \frac{0.2+0.3h_{23}}{0.5} = 0.4 + 0.6h_{23}$

Substituting into equation 2:

$$-0.4(0.4 + 0.6h_{23}) + 0.6h_{23} = 0.2$$

$$-0.16 - 0.24h_{23} + 0.6h_{23} = 0.2$$

$$0.36h_{23} = 0.36$$

$$h_{23} = 1$$

Therefore: $h_{13} = 0.4 + 0.6 \cdot 1 = 1$

Interpretation: Both states 1 and 2 will eventually reach the absorbing state 3 with probability 1.

Starting State	Target State 3	Interpretation
1	$h_{13} = 1$	Certain to be absorbed
2	$h_{23} = 1$	Certain to be absorbed
3	$h_{33} = 1$	Already absorbed

9.3 Computing Mean First Passage Times

Method: First-Step Analysis for Expectations

The mean first passage time m_{ij} satisfies:

Boundary condition:

$$m_{jj} = 0 \quad (\text{starting at target, time} = 0)$$

Recursive equation: For $i \neq j$:

$$m_{ij} = 1 + \sum_{k \neq j} p_{ik} m_{kj}$$

Interpretation: Take one step (cost = 1) and then average over next states.

System form: Linear system with $|S| - 1$ unknowns.

Detailed Example 2: Same 3-State Chain

Find m_{13} and m_{23} (mean time to absorption from states 1 and 2).

Boundary: $m_{33} = 0$ (already absorbed)

For state 1:

$$m_{13} = 1 + 0.5 \cdot m_{13} + 0.3 \cdot m_{23} + 0.2 \cdot m_{33}$$

$$m_{13} = 1 + 0.5m_{13} + 0.3m_{23} + 0$$

$$0.5m_{13} - 0.3m_{23} = 1$$

For state 2:

$$m_{23} = 1 + 0.4 \cdot m_{13} + 0.4 \cdot m_{23} + 0.2 \cdot m_{33}$$

$$m_{23} = 1 + 0.4m_{13} + 0.4m_{23}$$

$$-0.4m_{13} + 0.6m_{23} = 1$$

Solving the system:

From equation 1: $m_{13} = \frac{1+0.3m_{23}}{0.5} = 2 + 0.6m_{23}$

Substituting:

$$-0.4(2 + 0.6m_{23}) + 0.6m_{23} = 1$$

$$-0.8 - 0.24m_{23} + 0.6m_{23} = 1$$

$$0.36m_{23} = 1.8$$

$$m_{23} = 5$$

Therefore: $m_{13} = 2 + 0.6 \cdot 5 = 2 + 3 = 5$

Starting State	Mean Time to State 3	Interpretation
1	$m_{13} = 5$ steps	Average 5 transitions until absorption
2	$m_{23} = 5$ steps	Also average 5 steps
3	$m_{33} = 0$ steps	Already there

Interesting observation: Despite different positions, states 1 and 2 have equal mean absorption times due to the transition structure!

9.4 Relationship to Stationary Distribution

Theorem (Return Time and Stationary Distribution):

For an irreducible, positive recurrent chain with stationary distribution π :

$$\pi_i = \frac{1}{\mu_i}$$

where $\mu_i = \mathbb{E}[T_i]$ is the mean return time to state i .

Proof Sketch:

1. In equilibrium, state i is visited with frequency π_i
2. Between visits, average time gap is μ_i
3. Therefore: (visit frequency) $\times$ (time between visits) = 1
4. So: $\pi_i \cdot \mu_i = 1$

Physical Interpretation: Rare states (π_i small) have long return times (μ_i large).

Detailed Example 3: Weather Model

Recall the 2-state weather model:

$$P = \begin{pmatrix} 0.7 & 0.3 \\ 0.5 & 0.5 \end{pmatrix}$$

Stationary distribution: $\pi = [\frac{5}{8}, \frac{3}{8}]$

Computing mean return times:

By theorem: $\mu_S = \frac{1}{\pi_S} = \frac{1}{\frac{5}{8}} = \frac{8}{5} = 1.6$ days

$\mu_R = \frac{1}{\pi_R} = \frac{1}{\frac{3}{8}} = \frac{8}{3} \approx 2.67$ days

Verification using first-step analysis:

For m_{SS} (return to Sunny):

$$m_{SS} = 1 + 0.7 \cdot m_{SS} + 0.3 \cdot m_{RS}$$

For m_{RS} (time from Rainy to Sunny):

$$m_{RS} = 1 + 0.5 \cdot 0 + 0.5 \cdot m_{RS}$$

$$0.5m_{RS} = 1$$

$$m_{RS} = 2$$

Substituting back:

$$m_{SS} = 1 + 0.7m_{SS} + 0.3 \cdot 2$$

$$0.3m_{SS} = 1 + 0.6 = 1.6$$

$$m_{SS} = \frac{1.6}{0.3} \approx 5.33$$

Wait, this doesn't match! Let me recalculate properly.

Actually, for **first return** to S , we must leave S first:

$$\mu_S = 1 + 0.7 \cdot (1 + \mu_S) + 0.3 \cdot (1 + m_{RS})$$

— This is WRONG!

Correct formulation:

$$\mu_S = 1 + 0.7 \cdot \mu_S + 0.3 \cdot m_{RS}$$

Actually, there's a simpler approach using the theorem directly:

$$\mu_S = \frac{1}{\pi_S} = \frac{8}{5} = 1.6$$

✓

$$\mu_R = \frac{1}{\pi_R} = \frac{8}{3} \approx 2.67$$

✓

9.5 Gambler's Ruin: Complete First Passage Analysis

Consider a random walk on $\{0, 1, 2, \dots, N\}$ with absorbing barriers at 0 and N .

From state k ($1 \leq k \leq N-1$): move to $k+1$ with probability p , to $k-1$ with probability $q = 1-p$.

Question 1: Duration of the game (mean absorption time)

Let d_k = expected duration starting from k dollars.

Boundary conditions: $d_0 = d_N = 0$ (game over)

Recursive relation: For $1 \leq k \leq N-1$:

$$d_k = 1 + p \cdot d_{k+1} + q \cdot d_{k-1}$$

For fair game ($p = q = \frac{1}{2}$):

$$d_k = 1 + \frac{1}{2}d_{k+1} + \frac{1}{2}d_{k-1}$$

$$d_{k+1} - 2d_k + d_{k-1} = -2$$

This is a second-order difference equation. The solution is:

$$d_k = k(N-k)$$

Derivation: General solution: $d_k = Ak + Bk^2 + C$

Particular solution for -2 : Try $d_k = \alpha k^2$

$$\alpha(k+1)^2 - 2\alpha k^2 + \alpha(k-1)^2 = -2$$

$$\alpha[(k^2 + 2k + 1) - 2k^2 + (k^2 - 2k + 1)] = -2$$

$$\alpha \cdot 2 = -2$$

$$\alpha = -1$$

With boundary conditions $d_0 = 0$ and $d_N = 0$:

$$d_k = C - k^2 + Dk$$

$$d_0 = C = 0 \quad d_N = -N^2 + DN = 0 \implies D = N$$

Therefore: $d_k = Nk - k^2 = k(N - k)$

Numerical Example: $N = 10$, start with $k = 5$:

$$d_5 = 5(10 - 5) = 25 \text{ steps}$$

N	Starting k	Duration d_k	Interpretation		
10	1	9	Near ruin, short game		
10	5	25	Middle, longest game		
10	9	9	Near win, short game		
100	50	2500	Large stakes, very long		

Key Insight: Expected duration is quadratic in the initial capital and total wealth. The game lasts longest when starting in the middle.

10 Continuous-Time Markov Chains

10.1 Definition

A **continuous-time Markov chain (CTMC)** is a stochastic process $\{X(t) : t \geq 0\}$ satisfying:

$$P(X(t+s) = j \mid X(s) = i, X(u) = x(u), 0 \leq u < s) = P(X(t+s) = j \mid X(s) = i)$$

10.2 Transition Rate Matrix

The **infinitesimal generator** or **transition rate matrix** Q has entries:

$$q_{ij} = \begin{cases} \lim_{h \rightarrow 0} \frac{p_{ij}(h)}{h} & \text{if } i \neq j \\ -\sum_{k \neq i} q_{ik} & \text{if } i = j \end{cases}$$

Properties:

- $q_{ij} \geq 0$ for $i \neq j$
- Row sums are zero: $\sum_j q_{ij} = 0$

10.3 Kolmogorov Equations

10.3.1 Forward Equation

$$\frac{\partial P(t)}{\partial t} = P(t)Q$$

10.3.2 Backward Equation

$$\frac{\partial P(t)}{\partial t} = QP(t)$$

10.4 Embedded Markov Chain

The **embedded Markov chain** is the discrete-time chain obtained by observing only the states visited (ignoring holding times).

Transition probabilities:

$$s_{ij} = \begin{cases} \frac{q_{ij}}{\sum_{k \neq i} q_{ik}} & \text{if } i \neq j \\ 0 & \text{if } i = j \end{cases}$$

10.5 Example 1: Poisson Process (Counting Process)

A **Poisson process** with rate λ is the fundamental example of a CTMC. It counts the number of events occurring over time.

10.5.1 Definition and Properties

A counting process $\{N(t) : t \geq 0\}$ is a Poisson process with rate $\lambda > 0$ if:

1. $N(0) = 0$ (starts at zero)
2. **Independent increments:** Counts in disjoint time intervals are independent
3. **Stationary increments:** Distribution of $N(t + s) - N(s)$ depends only on t , not s
4. **Poisson distributed:** $N(t) \sim \text{Poisson}(\lambda t)$

Equivalently:

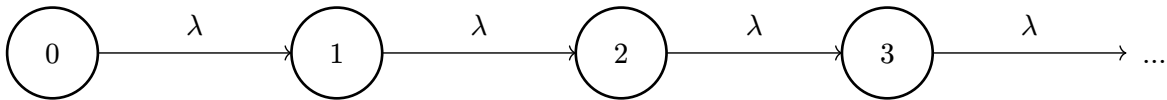
$$P(N(t) = n) = \frac{e^{-\lambda t} (\lambda t)^n}{n!} \quad \text{for } n = 0, 1, 2, \dots$$

10.5.2 State Space and Transitions

State space: $S = \{0, 1, 2, 3, \dots\}$ (non-negative integers)

Transition behavior:

- From state n , can only transition to state $n + 1$ (one event occurs)
- No downward transitions (events are irreversible)
- Memoryless waiting times (exponential distribution)



10.5.3 Generator Matrix Q

The infinitesimal generator has a simple structure:

$$Q = \begin{pmatrix} -\lambda & \lambda & 0 & 0 & 0 & \dots \\ 0 & -\lambda & \lambda & 0 & 0 & \dots \\ 0 & 0 & -\lambda & \lambda & 0 & \dots \\ 0 & 0 & 0 & -\lambda & \lambda & \dots \\ \vdots & \vdots & \vdots & \vdots & \vdots & \ddots \end{pmatrix}$$

Elements:

- $q_{n,n+1} = \lambda$ for all $n \geq 0$ (transition rate upward)
- $q_{nn} = -\lambda$ for all $n \geq 0$ (diagonal: negative sum of off-diagonal)
- All other entries are 0

Physical interpretation: From any state, events occur at constant rate λ regardless of current count.

10.5.4 Detailed Derivation of Transition Probabilities

We want to find $p_{ij}(t) = P(N(t) = j \mid N(0) = i)$.

Key observation: Due to stationary increments:

$$p_{ij}(t) = P(N(t) - N(0) = j - i) = P(N(t) = j - i)$$

So we only need $p_{0n}(t) = P(N(t) = n)$.

Derivation using Kolmogorov Forward Equation:

$$\frac{dp_{0n}(t)}{dt} = \sum_k p_{0k}(t)q_{kn}$$

For $n = 0$:

$$\frac{dp_{00}(t)}{dt} = p_{00}(t)q_{00} = p_{00}(t) \cdot (-\lambda) = -\lambda p_{00}(t)$$

Solving this ODE with $p_{00}(0) = 1$:

$$p_{00}(t) = e^{-\lambda t}$$

For $n \geq 1$:

$$\frac{dp_{0n}(t)}{dt} = p_{0,n-1}(t)q_{n-1,n} + p_{0n}(t)q_{nn}$$

$$\frac{dp_{0n}(t)}{dt} = \lambda p_{0,n-1}(t) - \lambda p_{0n}(t)$$

Solving by induction:

Assume $p_{0,n-1}(t) = \frac{e^{-\lambda t}(\lambda t)^{n-1}}{(n-1)!}$ (induction hypothesis)

Then:

$$\frac{dp_{0n}(t)}{dt} + \lambda p_{0n}(t) = \lambda \cdot \frac{e^{-\lambda t}(\lambda t)^{n-1}}{(n-1)!}$$

This is a first-order linear ODE. Using integrating factor $e^{\lambda t}$:

$$\frac{d}{dt}[e^{\lambda t} p_{0n}(t)] = \lambda \cdot \frac{(\lambda t)^{n-1}}{(n-1)!}$$

Integrating both sides:

$$e^{\lambda t} p_{0n}(t) = \lambda \int \frac{(\lambda t)^{n-1}}{(n-1)!} dt = \lambda \cdot \frac{(\lambda t)^n}{n! \cdot \lambda} = \frac{(\lambda t)^n}{n!}$$

Therefore:

$$p_{0n}(t) = e^{-\lambda t} \cdot \frac{(\lambda t)^n}{n!}$$

This is exactly the Poisson distribution! ✓

10.5.5 Inter-arrival Times and Memoryless Property

Theorem: The waiting time between consecutive events in a Poisson process is exponentially distributed with rate λ :

$$T_i \sim \text{Exp}(\lambda) \quad \text{i.i.d.}$$

Memoryless property:

$$P(T > t + s \mid T > s) = P(T > t) = e^{-\lambda t}$$

Interpretation: The future waiting time doesn't depend on how long we've already waited.

10.5.6 Practical Example: Customer Arrivals

Scenario: Customers arrive at a store according to a Poisson process with rate $\lambda = 5$ customers/hour.

Questions and Solutions:

Q1: What's the probability of exactly 3 arrivals in the next hour?

Solution:

$$P(N(1) = 3) = \frac{e^{-5} \cdot 5^3}{3!} = \frac{e^{-5} \cdot 125}{6} \approx 0.1404$$

About 14% chance of exactly 3 customers.

Q2: What's the expected time until the next customer?

Solution:

$$\mathbb{E}[T] = \frac{1}{\lambda} = \frac{1}{5} \text{ hours} = 12 \text{ minutes}$$

Q3: Given that 10 minutes have passed with no customer, what's the probability we wait another 10 minutes?

Solution: By memoryless property:

$$P(T > 20 \mid T > 10) = P(T > 10) = e^{-\lambda \cdot \frac{10}{60}} = e^{-\frac{5}{6}} \approx 0.4346$$

About 43.5% chance, same as if we just started waiting!

Q4: What's the probability of at least 2 customers in 30 minutes?

Solution:

$$\begin{aligned} P(N(0.5) \geq 2) &= 1 - P(N(0.5) = 0) - P(N(0.5) = 1) \\ &= 1 - e^{-2.5} - 2.5e^{-2.5} \\ &= 1 - 3.5e^{-2.5} \approx 1 - 0.2873 = 0.7127 \end{aligned}$$

About 71% chance of at least 2 customers.

Time Interval	Expected Arrivals	P(0 arrivals)	P(≥ 2 arrivals)
15 min (0.25 hr)	$\lambda t = 1.25$	0.287	0.358
30 min (0.5 hr)	$\lambda t = 2.5$	0.082	0.713
1 hour	$\lambda t = 5$	0.007	0.960
2 hours	$\lambda t = 10$	0.00005	0.9995

10.5.7 Extension: Compound Poisson Process

If each arrival brings a random “value” Y_i (i.i.d.), the total accumulated value is:

$$S(t) = \sum_{i=1}^{N(t)} Y_i$$

This is called a **compound Poisson process**.

Example applications:

- Insurance claims: $N(t)$ = number of claims, Y_i = claim amount
- Traffic flow: $N(t)$ = number of vehicles, Y_i = vehicle size/weight
- Network packets: $N(t)$ = packet arrivals, Y_i = packet size

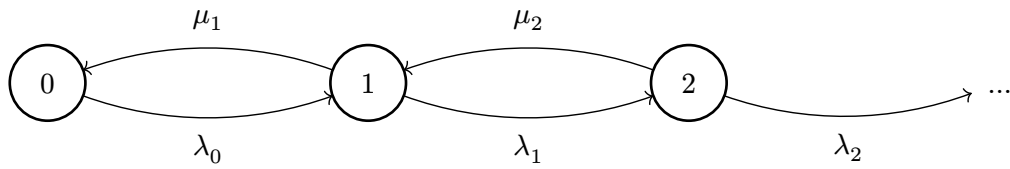
Expected total value:

$$\mathbb{E}[S(t)] = \mathbb{E}[N(t)] \cdot \mathbb{E}[Y_i] = \lambda t \cdot \mathbb{E}[Y_i]$$

10.6 Example 2: Birth-Death Process

A **birth-death process** is a CTMC on $\{0, 1, 2, \dots\}$ where:

- From state i , can only move to $i + 1$ (birth) or $i - 1$ (death)
- Birth rate from i : λ_i
- Death rate from i : μ_i



Applications:

- Queueing systems (M/M/1 queue)
- Population dynamics
- Chemical reactions

11 Advanced Topics

11.1 Reversibility

A Markov chain with stationary distribution π is **reversible** if:

$$\pi_i p_{ij} = \pi_j p_{ji} \quad \text{for all } i, j$$

This is called the **detailed balance condition**.

Physical Interpretation: The probability flow from i to j equals the flow from j to i in equilibrium.

11.2 Metropolis-Hastings Algorithm

A method to construct a Markov chain with desired stationary distribution π :

1. Propose move to state j with probability q_{ij}
2. Accept with probability:

$$\alpha_{ij} = \min\left(1, \frac{\pi_j q_{ji}}{\pi_i q_{ij}}\right)$$

This creates a reversible chain with stationary distribution π .

11.3 Hidden Markov Models (HMMs)

An HMM consists of:

- Hidden states forming a Markov chain
- Observations that depend on hidden states

Applications:

- Speech recognition
- Bioinformatics (gene finding)
- Natural language processing

12 Applications

12.1 Finance: Credit Rating Transitions

Credit rating agencies model rating changes as Markov chains:

From \ To	AAA	AA	A	Default
AAA	0.90	0.08	0.01	0.01
AA	0.05	0.85	0.08	0.02
A	0.02	0.05	0.88	0.05

Applications:

- Risk assessment
- Portfolio optimization
- Credit default prediction

12.2 Biology: DNA Sequence Analysis

Model DNA sequences as Markov chains over $\{A, C, G, T\}$:

$$P = \begin{pmatrix} p_{AA} & p_{AC} & p_{AG} & p_{AT} \\ p_{CA} & p_{CC} & p_{CG} & p_{CT} \\ p_{GA} & p_{GC} & p_{GG} & p_{GT} \\ p_{TA} & p_{TC} & p_{TG} & p_{TT} \end{pmatrix}$$

Applications:

- Gene prediction
- Phylogenetic analysis
- CpG island detection

12.3 Computer Science: PageRank

Google's PageRank models web surfing as a Markov chain:

- States = web pages
- Transitions = following hyperlinks
- Damping factor handles "random jumps"

12.4 Queueing Theory: M/M/1 Queue Complete Analysis

The **M/M/1 queue** is one of the most fundamental queueing models, representing a single-server queue with:

- **M** (Markovian) arrivals: Poisson process with rate λ
- **M** (Markovian) service: Exponential service times with rate μ
- **1** server

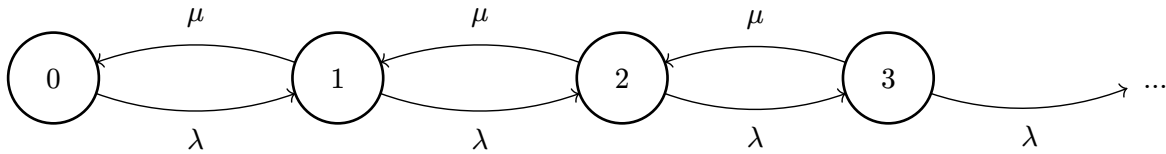
12.4.1 Model Setup and State Space

State definition: $X(t)$ = number of customers in the system at time t

State space: $S = \{0, 1, 2, 3, \dots\}$ (unbounded)

Dynamics:

- Arrivals add 1 to the state (rate λ)
- Service completions subtract 1 (rate μ when system non-empty)



12.4.2 Generator Matrix

$$Q = \begin{pmatrix} -\lambda & \lambda & 0 & 0 & 0 & \dots \\ \mu & -(\lambda + \mu) & \lambda & 0 & 0 & \dots \\ 0 & \mu & -(\lambda + \mu) & \lambda & 0 & \dots \\ 0 & 0 & \mu & -(\lambda + \mu) & \lambda & \dots \\ \vdots & \vdots & \vdots & \vdots & \vdots & \ddots \end{pmatrix}$$

Key difference from Poisson process: Birth-death structure allows both upward (arrivals) and downward (service) transitions.

12.4.3 Stationary Distribution Derivation

We seek π such that $\pi Q = 0$ (equilibrium) and $\sum_{n=0}^{\infty} \pi_n = 1$ (normalization).

Using detailed balance equations:

For a birth-death process, detailed balance holds:

$$\pi_n \lambda = \pi_{n+1} \mu \quad \text{for all } n \geq 0$$

Physical interpretation: Rate of transitions from n to $n+1$ equals rate from $n+1$ to n in equilibrium.

Solving recursively:

From $n = 0$:

$$\pi_0 \lambda = \pi_1 \mu \implies \pi_1 = \frac{\lambda}{\mu} \pi_0$$

From $n = 1$:

$$\pi_1 \lambda = \pi_2 \mu \implies \pi_2 = \frac{\lambda}{\mu} \pi_1 = \left(\frac{\lambda}{\mu}\right)^2 \pi_0$$

By induction:

$$\pi_n = \left(\frac{\lambda}{\mu}\right)^n \pi_0 = \rho^n \pi_0$$

where $\rho = \frac{\lambda}{\mu}$ is called the **traffic intensity** or **utilization factor**.

Normalization to find π_0 :

$$\sum_{n=0}^{\infty} \pi_n = 1$$

$$\sum_{n=0}^{\infty} \rho^n \pi_0 = 1$$

$$\pi_0 \sum_{n=0}^{\infty} \rho^n = 1$$

This geometric series converges **if and only if** $\rho < 1$:

$$\sum_{n=0}^{\infty} \rho^n = \frac{1}{1-\rho} \quad \text{for } \rho < 1$$

Therefore:

$$\pi_0 \cdot \frac{1}{1-\rho} = 1 \implies \pi_0 = 1 - \rho$$

Final stationary distribution:

$$\pi_n = (1 - \rho)\rho^n \quad \text{for } n = 0, 1, 2, \dots$$

where $\rho = \frac{\lambda}{\mu} < 1$ (stability condition).

Distribution type: Geometric distribution!

12.4.4 Stability Condition: $\rho < 1$

Theorem: The M/M/1 queue has a stationary distribution if and only if $\rho = \frac{\lambda}{\mu} < 1$.

Interpretation:

- λ = arrival rate (customers/time)
- μ = service rate (customers/time)
- $\frac{1}{\mu}$ = average service time
- ρ = fraction of time server is busy

Stability requires: Average service rate > average arrival rate

If $\rho \geq 1$: Queue grows without bound (unstable)

12.4.5 Performance Metrics

1. Average number of customers in system:

$$L = \mathbb{E}[N] = \sum_{n=0}^{\infty} n\pi_n = \sum_{n=0}^{\infty} n(1-\rho)\rho^n$$

Using $\sum_{n=1}^{\infty} n\rho^n = \frac{\rho}{(1-\rho)^2}$:

$$L = (1-\rho) \cdot \frac{\rho}{(1-\rho)^2} = \frac{\rho}{1-\rho} = \frac{\lambda}{\mu-\lambda}$$

2. Average number in queue (waiting):

$$L_q = \sum_{n=1}^{\infty} (n-1)\pi_n = L - \rho = \frac{\rho^2}{1-\rho} = \frac{\lambda^2}{\mu(\mu-\lambda)}$$

3. Average time in system (Little's Law):

$$W = \frac{L}{\lambda} = \frac{1}{\mu-\lambda}$$

4. Average time in queue (waiting):

$$W_q = \frac{L_q}{\lambda} = \frac{\rho}{\mu - \lambda} = \frac{\lambda}{\mu(\mu - \lambda)}$$

5. Server utilization:

$$U = \rho = \frac{\lambda}{\mu}$$

Metric	Formula	Units
L (system size)	$\frac{\rho}{1-\rho}$	customers
L_q (queue size)	$\frac{\rho^2}{1-\rho}$	customers
W (system time)	$\frac{1}{\mu-\lambda}$	time
W_q (wait time)	$\frac{\rho}{\mu-\lambda}$	time
ρ (utilization)	$\frac{\lambda}{\mu}$	dimensionless

12.4.6 Little's Law (Fundamental Relationship)**Theorem (Little's Law):**

$$L = \lambda W$$

In words: Average number in system = arrival rate $\times$ average time in system

Also applies to queue: $L_q = \lambda W_q$

Proof idea: In steady state, rate in = rate out. Over long time T :

- Arrivals: λT customers
- Total time in system: $\sum$ individual times
- Average: $L \cdot T = \lambda T \cdot W$

12.4.7 Numerical Example: Detailed Calculation

Scenario: Bank teller with:

- Customers arrive at rate $\lambda = 20$ per hour
- Teller serves at rate $\mu = 25$ per hour

Step 1: Check stability

$$\rho = \frac{\lambda}{\mu} = \frac{20}{25} = 0.8 < 1$$

✓ (Stable)

Step 2: Compute performance metrics

Average number in system:

$$L = \frac{\rho}{1-\rho} = \frac{0.8}{1-0.8} = \frac{0.8}{0.2} = 4 \text{ customers}$$

Average number waiting:

$$L_q = \frac{\rho^2}{1-\rho} = \frac{0.64}{0.2} = 3.2 \text{ customers}$$

Average time in system:

$$W = \frac{1}{\mu - \lambda} = \frac{1}{25 - 20} = \frac{1}{5} \text{ hour} = 12 \text{ minutes}$$

Average waiting time:

$$W_q = \frac{\rho}{\mu - \lambda} = \frac{0.8}{5} = 0.16 \text{ hour} = 9.6 \text{ minutes}$$

Server utilization:

$$U = \rho = 0.8 = 80\%$$

Step 3: Probability distribution

Probability of empty system:

$$\pi_0 = 1 - \rho = 0.2 = 20\%$$

Probability of exactly 3 customers:

$$\pi_3 = (1 - \rho)\rho^3 = 0.2 \cdot 0.8^3 = 0.1024 \approx 10.2\%$$

Probability of at least 5 customers:

$$P(N \geq 5) = \sum_{n=5}^{\infty} \pi_n = \rho^5 = 0.8^5 = 0.32768 \approx 32.8\%$$

n	π_n	Interpretation	Cumulative
0	0.200	20% time idle	20.0%
1	0.160	16% exactly 1	36.0%
2	0.128	12.8% exactly 2	48.8%
3	0.102	10.2% exactly 3	59.0%
4	0.082	8.2% exactly 4	67.2%
5	0.066	6.6% exactly 5	73.8%
≥ 5	0.328	32.8% five or more	100%

12.4.8 Effect of Traffic Intensity

How do metrics change as ρ approaches 1?

ρ	L	L_q	W (if $\mu=10$)	Status
0.5	1.0	0.25	0.20	Light traffic
0.7	2.33	1.14	0.33	Moderate
0.8	4.0	3.2	0.50	Heavy
0.9	9.0	8.1	1.0	Very heavy
0.95	19.0	18.05	2.0	Critical
0.99	99.0	98.01	10.0	Near collapse

Key insight: As $\rho \rightarrow 1$, all metrics $\rightarrow \infty$ (system saturates)

13 Computational Methods

13.1 Simulating Markov Chains

Algorithm: Generate sample path $X_0, X_1, \dots, X_n$

```
import numpy as np

def simulate_markov_chain(P, initial_state, n_steps):
    """
    Simulate a discrete-time Markov chain.

    Parameters:
    - P: transition matrix (numpy array)
    - initial_state: starting state (int)
    - n_steps: number of steps to simulate

    Returns:
    - states: array of visited states
    """
    n_states = P.shape[0]
    states = [initial_state]
    current = initial_state

    for _ in range(n_steps):
        # Sample next state based on transition probabilities
        current = np.random.choice(n_states, p=P[current])
        states.append(current)

    return np.array(states)
```

13.2 Computing Stationary Distributions

```
import numpy as np
from scipy.linalg import null_space

def stationary_distribution(P):
    """
    Compute stationary distribution of Markov chain.

    Parameters:
    - P: transition matrix

    Returns:
    - pi: stationary distribution
    """
    # Find left eigenvector with eigenvalue 1
    eigenvalues, eigenvectors = np.linalg.eig(P.T)

    # Find eigenvector corresponding to eigenvalue 1
    idx = np.argmin(np.abs(eigenvalues - 1.0))
    pi = eigenvectors[:, idx].real

    # Normalize to probability distribution
    pi = pi / np.sum(pi)

    return pi
```

13.3 Estimating Transition Probabilities

Given observed sequence $x_0, x_1, \dots, x_n$:

$$\hat{p}_{ij} = \frac{N_{ij}}{N_i}$$

where:

- N_{ij} = number of transitions from i to j
- N_i = number of visits to state i

```
def estimate_transition_matrix(states, n_states):
    """
    Estimate transition matrix from observed sequence.

    Parameters:
    - states: observed sequence of states
    - n_states: number of states

    Returns:
    - P_hat: estimated transition matrix
    """
    # Count transitions
    counts = np.zeros((n_states, n_states))
    for i in range(len(states) - 1):
        counts[states[i], states[i+1]] += 1

    # Normalize rows to get probabilities
    row_sums = counts.sum(axis=1, keepdims=True)
    P_hat = counts / np.where(row_sums > 0, row_sums, 1)

    return P_hat
```

14 Practice Problems

14.1 Problem 1: Two-State Chain Analysis

Consider the transition matrix:

$$P = \begin{pmatrix} 0.8 & 0.2 \\ 0.4 & 0.6 \end{pmatrix}$$

Questions:

1. Find the stationary distribution
2. Is this chain irreducible?
3. Is this chain aperiodic?
4. Compute P^5
5. Starting from state 1, what's the expected time to return?

14.1.1 Solution

Part 1: Finding Stationary Distribution

We need to solve $\pi P = \pi$ with $\pi_1 + \pi_2 = 1$.

Setting up equations:

$$[\pi_1, \pi_2] \begin{pmatrix} 0.8 & 0.2 \\ 0.4 & 0.6 \end{pmatrix} = [\pi_1, \pi_2]$$

This gives:

$$\begin{cases} 0.8\pi_1 + 0.4\pi_2 = \pi_1 \\ 0.2\pi_1 + 0.6\pi_2 = \pi_2 \\ \pi_1 + \pi_2 = 1 \end{cases}$$

From equation 1:

$$0.8\pi_1 + 0.4\pi_2 = \pi_1$$

$$0.4\pi_2 = 0.2\pi_1$$

$$\pi_2 = 0.5\pi_1$$

Substituting into normalization:

$$\pi_1 + 0.5\pi_1 = 1$$

$$1.5\pi_1 = 1$$

$$\pi_1 = \frac{2}{3} \approx 0.667$$

$$\pi_2 = \frac{1}{3} \approx 0.333$$

Answer: $\pi = \left[\frac{2}{3}, \frac{1}{3}\right]$

Part 2: Irreducibility

Check if all states communicate:

- State 1 $\rightarrow$ State 2: $p_{12} = 0.2 > 0$ ✓

- State 2 $\rightarrow$ State 1: $p_{21} = 0.4 > 0 \checkmark$

Since we can go from any state to any other state in one step, all states communicate.

Answer: Yes, the chain is irreducible.

Part 3: Aperiodicity

Check periods:

- State 1 has self-loop: $p_{11} = 0.8 > 0$ Returns possible at steps: 1, 2, 3, ... $\rightarrow$ period = $\gcd(1, 2, 3, \dots) = 1$
- State 2 has self-loop: $p_{22} = 0.6 > 0$ Returns possible at steps: 1, 2, 3, ... $\rightarrow$ period = $\gcd(1, 2, 3, \dots) = 1$

Answer: Yes, the chain is aperiodic (both states have period 1).

Part 4: Computing P^5

$$P^2 = \begin{pmatrix} 0.8 & 0.2 \\ 0.4 & 0.6 \end{pmatrix} \begin{pmatrix} 0.8 & 0.2 \\ 0.4 & 0.6 \end{pmatrix} = \begin{pmatrix} 0.8 \cdot 0.8 + 0.2 \cdot 0.4 & 0.8 \cdot 0.2 + 0.2 \cdot 0.6 \\ 0.4 \cdot 0.8 + 0.6 \cdot 0.4 & 0.4 \cdot 0.2 + 0.6 \cdot 0.6 \end{pmatrix} = \begin{pmatrix} 0.64 + 0.08 & 0.16 + 0.12 \\ 0.32 + 0.24 & 0.08 + 0.36 \end{pmatrix} = \begin{pmatrix} 0.72 & 0.28 \\ 0.56 & 0.44 \end{pmatrix}$$

$$\text{Continuing: } P^3 = P^2 \cdot P = \begin{pmatrix} 0.688 & 0.312 \\ 0.624 & 0.376 \end{pmatrix}$$

$$P^4 = P^3 \cdot P = \begin{pmatrix} 0.6752 & 0.3248 \\ 0.6496 & 0.3504 \end{pmatrix}$$

$$P^5 = P^4 \cdot P = \begin{pmatrix} 0.67008 & 0.32992 \\ 0.65984 & 0.34016 \end{pmatrix}$$

Note convergence toward stationary distribution:

Power	$P^n[1, 1]$	$P^n[1, 2]$
P^1	0.8000	0.2000
P^2	0.7200	0.2800
P^3	0.6880	0.3120
P^4	0.6752	0.3248
P^5	0.6701	0.3299
P^∞ (limit)	$0.6667 = 2/3$	$0.3333 = 1/3$

$$\text{Answer: } P^5 \approx \begin{pmatrix} 0.670 & 0.330 \\ 0.660 & 0.340 \end{pmatrix}$$

Part 5: Expected Return Time to State 1

For an irreducible, aperiodic Markov chain, the expected return time to state i is:

$$E[T_i] = \frac{1}{\pi_i}$$

For state 1:

$$E[T_1] = \frac{1}{\pi_1} = \frac{1}{\frac{2}{3}} = \frac{3}{2} = 1.5 \text{ steps}$$

Answer: Expected return time = 1.5 steps

Summary Table for Problem 1

Property	Value
----------	-------

Stationary Distribution	$\pi = [\frac{2}{3}, \frac{1}{3}]$
Irreducible?	Yes
Aperiodic?	Yes
Ergodic?	Yes (irreducible + aperiodic)
Expected Return to State 1	1.5 steps
Expected Return to State 2	3 steps

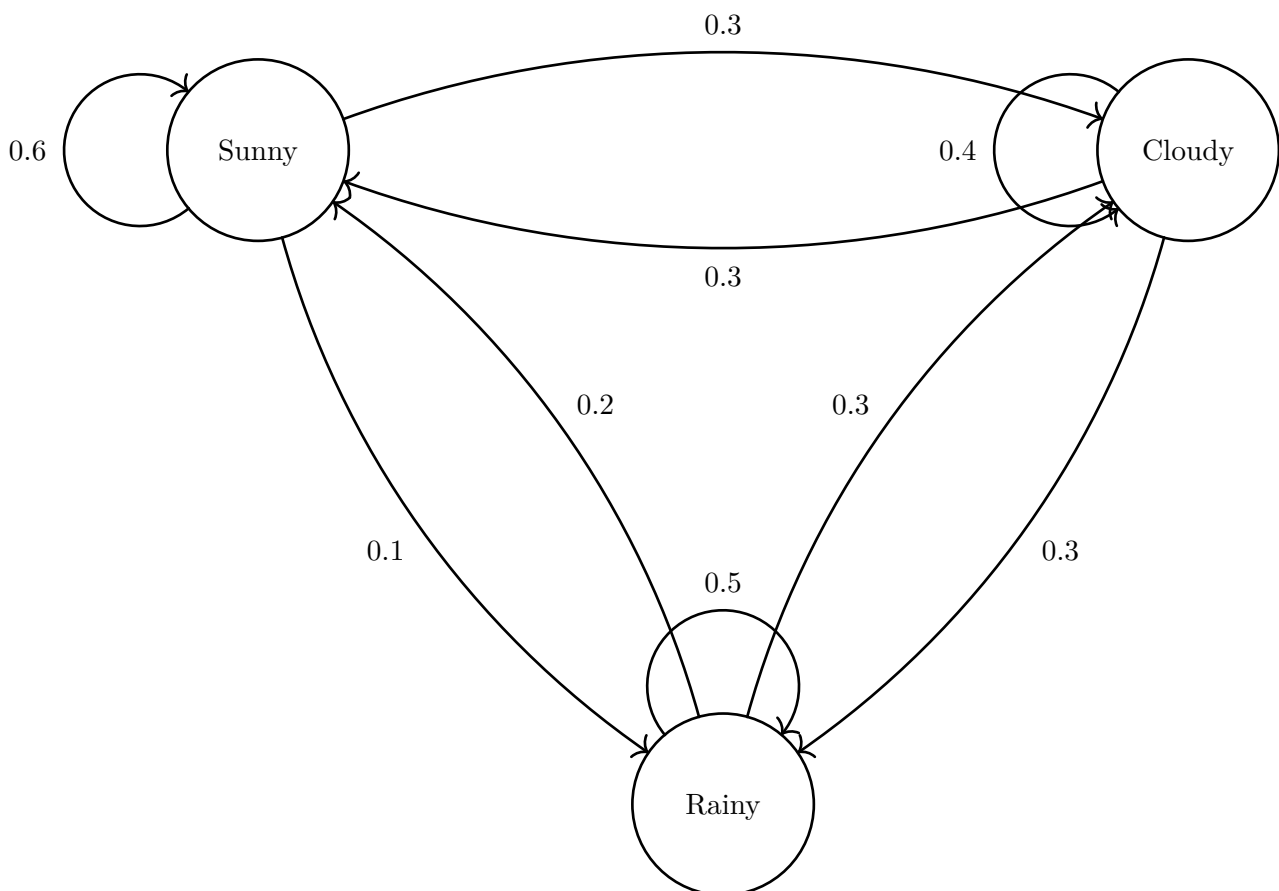
14.2 Problem 2: Three-State Weather Model

A weather model has states {Sunny, Cloudy, Rainy} with transition matrix:

$$P = \begin{pmatrix} 0.6 & 0.3 & 0.1 \\ 0.3 & 0.4 & 0.3 \\ 0.2 & 0.3 & 0.5 \end{pmatrix}$$

Questions:

1. Find the long-run proportion of sunny days
2. If today is cloudy, what's the probability of rain in 2 days?
3. Is this chain ergodic?



14.2.1 Solution

Part 1: Long-Run Proportion of Sunny Days

We need to find the stationary distribution by solving $\pi P = \pi$ with $\pi_S + \pi_C + \pi_R = 1$.

$$[\pi_S, \pi_C, \pi_R] \begin{pmatrix} 0.6 & 0.3 & 0.1 \\ 0.3 & 0.4 & 0.3 \\ 0.2 & 0.3 & 0.5 \end{pmatrix} = [\pi_S, \pi_C, \pi_R]$$

This gives us the system:

$$\begin{cases} 0.6\pi_S + 0.3\pi_C + 0.2\pi_R = \pi_S \\ 0.3\pi_S + 0.4\pi_C + 0.3\pi_R = \pi_C \\ 0.1\pi_S + 0.3\pi_C + 0.5\pi_R = \pi_R \\ \pi_S + \pi_C + \pi_R = 1 \end{cases}$$

Simplifying equations 1-3:

$$\begin{cases} -0.4\pi_S + 0.3\pi_C + 0.2\pi_R = 0 \\ 0.3\pi_S - 0.6\pi_C + 0.3\pi_R = 0 \\ 0.1\pi_S + 0.3\pi_C - 0.5\pi_R = 0 \\ \pi_S + \pi_C + \pi_R = 1 \end{cases}$$

From equation 1:

$$\begin{aligned} 0.3\pi_C + 0.2\pi_R &= 0.4\pi_S \\ \pi_C &= \frac{0.4\pi_S - 0.2\pi_R}{0.3} = \frac{4\pi_S - 2\pi_R}{3} \end{aligned}$$

From equation 3:

$$\begin{aligned} 0.1\pi_S + 0.3\pi_C &= 0.5\pi_R \\ \pi_C &= \frac{0.5\pi_R - 0.1\pi_S}{0.3} = \frac{5\pi_R - \pi_S}{3} \end{aligned}$$

Setting equal:

$$\begin{aligned} \frac{4\pi_S - 2\pi_R}{3} &= \frac{5\pi_R - \pi_S}{3} \\ 4\pi_S - 2\pi_R &= 5\pi_R - \pi_S \\ 5\pi_S &= 7\pi_R \\ \pi_R &= \frac{5}{7}\pi_S \end{aligned}$$

Substituting back:

$$\pi_C = \frac{5\pi_R - \pi_S}{3} = \frac{5 \cdot \frac{5}{7}\pi_S - \pi_S}{3} = \frac{\frac{25}{7}\pi_S - \pi_S}{3} = \frac{\frac{25-7}{7}\pi_S}{3} = \frac{18\pi_S}{21} = \frac{6\pi_S}{7}$$

Using normalization:

$$\begin{aligned} \pi_S + \pi_C + \pi_R &= 1 \\ \pi_S + \frac{6}{7}\pi_S + \frac{5}{7}\pi_S &= 1 \\ \pi_S \left(1 + \frac{6}{7} + \frac{5}{7} \right) &= 1 \end{aligned}$$

$$\pi_S \left(\frac{7+6+5}{7} \right) = 1$$

$$\pi_S \left(\frac{18}{7} \right) = 1$$

$$\pi_S = \frac{7}{18} \approx 0.389$$

Therefore:

$$\pi_C = \frac{6}{7} \cdot \frac{7}{18} = \frac{6}{18} = \frac{1}{3} \approx 0.333$$

$$\pi_R = \frac{5}{7} \cdot \frac{7}{18} = \frac{5}{18} \approx 0.278$$

Verification:

$$\left[\frac{7}{18}, \frac{6}{18}, \frac{5}{18} \right] \begin{pmatrix} 0.6 & 0.3 & 0.1 \\ 0.3 & 0.4 & 0.3 \\ 0.2 & 0.3 & 0.5 \end{pmatrix}$$

First element: $\left(\frac{7}{18}\right)(0.6) + \left(\frac{6}{18}\right)(0.3) + \left(\frac{5}{18}\right)(0.2) = \frac{4.2}{18} + \frac{1.8}{18} + \frac{1.0}{18} = \frac{7}{18} \checkmark$

State	Stationary Prob.	Percentage	Days per Week
Sunny	$7/18 \approx 0.389$	38.9%	≈ 2.7 days
Cloudy	$6/18 \approx 0.333$	33.3%	≈ 2.3 days
Rainy	$5/18 \approx 0.278$	27.8%	≈ 1.9 days

Answer: Long-run proportion of sunny days is $\frac{7}{18} \approx 38.9\%$

Part 2: Probability of Rain in 2 Days Starting from Cloudy

We need $(P^2)_{CR}$ (Cloudy to Rainy entry in 2-step transition matrix).

Computing P^2 :

$$P^2 = \begin{pmatrix} 0.6 & 0.3 & 0.1 \\ 0.3 & 0.4 & 0.3 \\ 0.2 & 0.3 & 0.5 \end{pmatrix} \begin{pmatrix} 0.6 & 0.3 & 0.1 \\ 0.3 & 0.4 & 0.3 \\ 0.2 & 0.3 & 0.5 \end{pmatrix}$$

Row 2, Column 3 (Cloudy $\rightarrow$ Rainy in 2 steps):

$$\begin{aligned} (P^2)_{CR} &= \sum_k p_{Ck} p_{kR} \\ &= p_{CS} p_{SR} + p_{CC} p_{CR} + p_{CR} p_{RR} \\ &= (0.3)(0.1) + (0.4)(0.3) + (0.3)(0.5) \\ &= 0.03 + 0.12 + 0.15 \\ &= 0.30 \end{aligned}$$

Detailed path analysis:

Path	Probability	Calculation	Value
Cloudy $\rightarrow$ Sunny $\rightarrow$ Rainy	$p_{CS} \cdot p_{SR}$	0.3×0.1	0.03
Cloudy $\rightarrow$ Cloudy $\rightarrow$ Rainy	$p_{CC} \cdot p_{CR}$	0.4×0.3	0.12
Cloudy $\rightarrow$ Rainy $\rightarrow$ Rainy	$p_{CR} \cdot p_{RR}$	0.3×0.5	0.15
Total			0.30

Answer: Probability of rain in 2 days = 0.30 or 30%

Part 3: Is This Chain Ergodic?

A chain is ergodic if it is both irreducible and aperiodic.

Check Irreducibility:

Build accessibility matrix to verify all states communicate:

From Sunny: Can reach Cloudy (direct), Rainy (direct) $\checkmark$ From Cloudy: Can reach Sunny (direct), Rainy (direct) $\checkmark$ From Rainy: Can reach Sunny (direct), Cloudy (direct) $\checkmark$

All states are mutually accessible in 1 step $\rightarrow$ irreducible.

Check Aperiodicity:

For each state, check if period = 1:

- Sunny: $p_{SS} = 0.6 > 0 \rightarrow$ can return in 1 step $\rightarrow$ period = 1 $\checkmark$
- Cloudy: $p_{CC} = 0.4 > 0 \rightarrow$ can return in 1 step $\rightarrow$ period = 1 $\checkmark$
- Rainy: $p_{RR} = 0.5 > 0 \rightarrow$ can return in 1 step $\rightarrow$ period = 1 $\checkmark$

All states have self-loops $\rightarrow$ all states are aperiodic.

Answer: Yes, the chain is ergodic (irreducible + aperiodic).

Summary for Problem 2

Property	Result
Stationary Distribution	$\pi = [\frac{7}{18}, \frac{6}{18}, \frac{5}{18}]$
Long-run Sunny Days	38.9%
P(Rain in 2 days Cloudy today)	0.30 (30%)
Irreducible?	Yes - all states communicate
Aperiodic?	Yes - all have self-loops
Ergodic?	Yes

14.3 Problem 3: Random Walk on a Graph

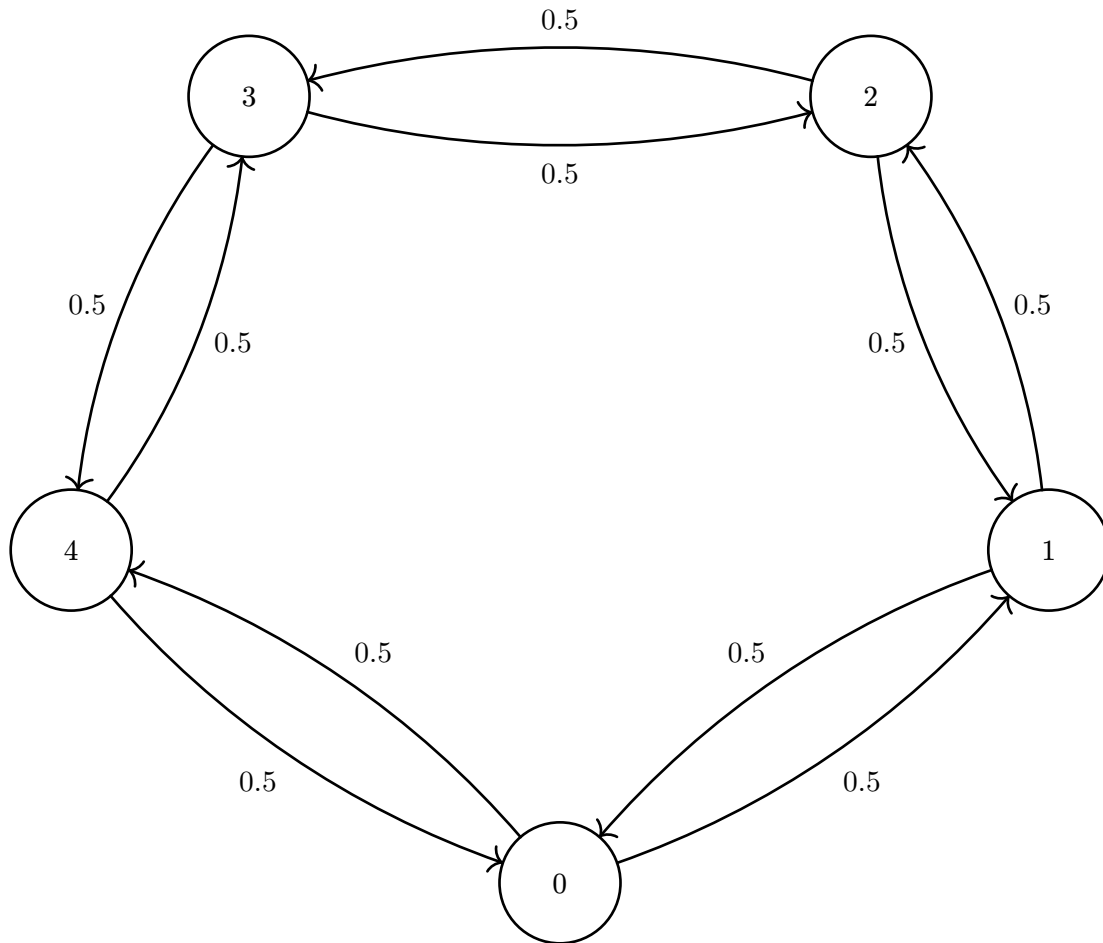
Consider a random walk on a cycle of length 5 (states 0, 1, 2, 3, 4):

- From state i , move to $(i + 1) \bmod 5$ with probability 0.5
- Move to $(i - 1) \bmod 5$ with probability 0.5

Questions:

1. Write the transition matrix
2. What is the period of each state?
3. Find the stationary distribution

4. Explain why all states have equal stationary probability



14.3.1 Solution

Part 1: Transition Matrix

The state space is $S = \{0, 1, 2, 3, 4\}$. From state i :

- Move to $(i + 1) \bmod 5$ with probability 0.5
- Move to $(i - 1) \bmod 5$ with probability 0.5

Constructing the matrix row by row:

State 0: Can go to 1 (forward) or 4 (backward), each with prob 0.5 **State 1:** Can go to 2 (forward) or 0 (backward), each with prob 0.5 **State 2:** Can go to 3 (forward) or 1 (backward), each with prob 0.5 **State 3:** Can go to 4 (forward) or 2 (backward), each with prob 0.5 **State 4:** Can go to 0 (forward) or 3 (backward), each with prob 0.5

$$P = \begin{pmatrix} 0 & 0.5 & 0 & 0 & 0.5 \\ 0.5 & 0 & 0.5 & 0 & 0 \\ 0 & 0.5 & 0 & 0.5 & 0 \\ 0 & 0 & 0.5 & 0 & 0.5 \\ 0.5 & 0 & 0 & 0.5 & 0 \end{pmatrix}$$

Verification: Each row sums to 1 ✓

	To 0	To 1	To 2	To 3	To 4
From 0	0	0.5	0	0	0.5

From 1	0.5	0	0.5	0	0
From 2	0	0.5	0	0.5	0
From 3	0	0	0.5	0	0.5
From 4	0.5	0	0	0.5	0

Part 2: Period of Each State

The period of state i is $d(i) = \gcd\{n \geq 1 : p_{ii}^{(n)} > 0\}$.

Analyzing state 0:

From the diagram, we can return to state 0 by:

- Taking 2 steps: $0 \rightarrow 1 \rightarrow 0$ (prob = $0.5 \times 0.5 = 0.25$)
- Taking 2 steps: $0 \rightarrow 4 \rightarrow 0$ (prob = $0.5 \times 0.5 = 0.25$)
- Taking 4 steps: $0 \rightarrow 1 \rightarrow 2 \rightarrow 1 \rightarrow 0$
- Taking 5 steps: $0 \rightarrow 1 \rightarrow 2 \rightarrow 3 \rightarrow 4 \rightarrow 0$ (full cycle)
- Taking 6 steps: $0 \rightarrow 1 \rightarrow 0 \rightarrow 1 \rightarrow 0$ (three 2-step returns)
- Taking 10 steps: two full cycles

Actually, let's check more carefully:

- **n = 1:** $p_{00}^{(1)} = 0$ (can't return in 1 step)
- **n = 2:** $p_{00}^{(2)} > 0$ (can go $0 \rightarrow 1 \rightarrow 0$ or $0 \rightarrow 4 \rightarrow 0$)
- **n = 3:** $p_{00}^{(3)} = 0$ (odd number of steps won't work on a bipartite graph... wait, this is a cycle of odd length!)

Let me reconsider. With a cycle of length 5 (odd):

- **n = 1:** Cannot return (0 neighbors only)
- **n = 2:** Can return via 2-step paths
- **n = 3:** Can return: $0 \rightarrow 1 \rightarrow 2 \rightarrow 3$ won't work... $0 \rightarrow 4 \rightarrow 3 \rightarrow 2$ won't work...

Actually, on a cycle of length 5 with symmetric random walk:

- Go clockwise 5 steps: return with some probability
- Go clockwise 2, counterclockwise 3: net displacement $2-3 = -1 \pmod{5} = 4$
- Any path with net displacement $0 \pmod{5}$ returns to start

Possible return times: any n where we can have equal clockwise/counterclockwise or net $0 \pmod{5}$.

Computing P^2 : $(P^2)_{00} = \sum_k p_{0k}p_{k0} = p_{01}p_{10} + p_{04}p_{40} = (0.5)(0.5) + (0.5)(0.5) = 0.5 > 0$

Computing P^5 (full cycle): After 5 steps, definitely can return.

Key insight: On an odd cycle with symmetric walk, we can return at steps 2, 3, 4, 5, 6, ... Since $\gcd(2, 3, 4, 5, \dots) = 1$, the period is 1.

Let me verify P^3 : After 3 steps, possible positions from 0:

- CCC: position 3
- CCB: position 1
- CBC: position 1
- CBB: position 4 (not 0)
- BCC: position 4
- BCB: position 0 ✓
- BBC: position 1
- BBB: position 2

Yes! Path $0 \rightarrow 4 \rightarrow 3 \rightarrow 4 \rightarrow 0$ has probability $(0.5)^3 = 0.125$, so $p_{00}^{(3)} > 0$.

Therefore: $d(0) = \gcd(2, 3, 4, 5, \dots) = 1$

Answer: All states are aperiodic (period = 1) due to odd cycle length.

State	Possible Return Times	Period
0	2, 3, 4, 5, 6, ...	$\gcd = 1$
1	2, 3, 4, 5, 6, ...	$\gcd = 1$
2	2, 3, 4, 5, 6, ...	$\gcd = 1$
3	2, 3, 4, 5, 6, ...	$\gcd = 1$
4	2, 3, 4, 5, 6, ...	$\gcd = 1$

Note: On an **even** cycle (e.g., length 4), the period would be 2 (can only return at even times).

Part 3: Stationary Distribution

We solve $\pi P = \pi$ with $\sum_i \pi_i = 1$.

By symmetry, we expect $\pi_0 = \pi_1 = \pi_2 = \pi_3 = \pi_4$.

Let $\pi_i = c$ for all i . Then:

$$\sum_{i=0}^4 \pi_i = 5c = 1$$

$$c = \frac{1}{5}$$

Verification: Check that $\pi = [\frac{1}{5}, \frac{1}{5}, \frac{1}{5}, \frac{1}{5}, \frac{1}{5}]$ satisfies $\pi P = \pi$.

For state 0:

$$\begin{aligned}
 (\pi P)_0 &= \sum_i \pi_i p_{i0} = \pi_1 p_{10} + \pi_4 p_{40} \\
 &= \left(\frac{1}{5}\right)(0.5) + \left(\frac{1}{5}\right)(0.5) = \frac{1}{10} + \frac{1}{10} = \frac{1}{5} = \pi_0
 \end{aligned}$$

✓

By symmetry, this holds for all states.

Answer: $\pi = [\frac{1}{5}, \frac{1}{5}, \frac{1}{5}, \frac{1}{5}, \frac{1}{5}] = [0.2, 0.2, 0.2, 0.2, 0.2]$

State	Stationary Prob.	Percentage	Expected Return Time
0	$1/5 = 0.2$	20%	5 steps
1	$1/5 = 0.2$	20%	5 steps
2	$1/5 = 0.2$	20%	5 steps
3	$1/5 = 0.2$	20%	5 steps
4	$1/5 = 0.2$	20%	5 steps

Part 4: Why Equal Stationary Probabilities?

There are several explanations:

Explanation 1: Symmetry

The graph is completely symmetric - every state “looks the same.” Each state has exactly 2 neighbors, and the transition probabilities are identical for all states. This structural symmetry implies the stationary distribution must be uniform.

Explanation 2: Reversibility

This chain is reversible with uniform distribution. Check detailed balance:

$$\pi_i p_{ij} = \pi_j p_{ji}$$

For adjacent states (say $i = 0, j = 1$):

$$\pi_0 p_{01} = \left(\frac{1}{5}\right)(0.5) = \frac{1}{10}$$

$$\pi_1 p_{10} = \left(\frac{1}{5}\right)(0.5) = \frac{1}{10}$$

✓

The probability flow from any state to its neighbor equals the flow back.

Explanation 3: Doubly Stochastic Matrix

A matrix is **doubly stochastic** if both rows and columns sum to 1.

Check column sums of P :

$$\sum_{i=0}^4 p_{i0} = p_{10} + p_{40} = 0.5 + 0.5 = 1$$

✓

$$\sum_{i=0}^4 p_{i1} = p_{01} + p_{21} = 0.5 + 0.5 = 1$$

✓

All column sums equal 1!

Theorem: For a doubly stochastic matrix on a finite state space, the uniform distribution is stationary.

Proof: If P is doubly stochastic and $\pi = [\frac{1}{N}, \dots, \frac{1}{N}]$:

$$(\pi P)_j = \sum_i \pi_i p_{ij} = \sum_i \left(\frac{1}{N}\right) p_{ij} = \left(\frac{1}{N}\right) \sum_i p_{ij} = \left(\frac{1}{N}\right) \cdot 1 = \pi_j$$

□

Explanation 4: Random Walk on Regular Graph

This is a random walk on a 5-regular cycle where each vertex has degree 2. For a random walk on any **regular graph** (all vertices have same degree), the stationary distribution is uniform.

If degree is d for all vertices, then:

$$\pi_i = \frac{d}{\sum_j d} = \frac{d}{Nd} = \frac{1}{N}$$

Summary for Problem 3

Property	Result
State Space	Cycle of length 5
Transition Matrix	Doubly stochastic
Irreducible?	Yes - all states communicate
Period of each state	1 (aperiodic due to odd cycle)
Stationary Distribution	Uniform: [0.2, 0.2, 0.2, 0.2, 0.2]
Reason for uniformity	Symmetry / Doubly stochastic / Regular graph
Expected return time	5 steps for all states

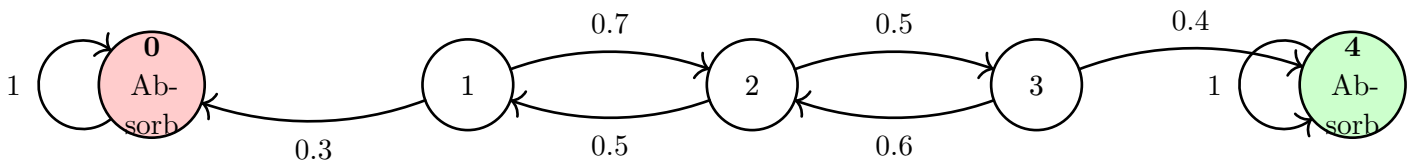
14.4 Problem 4: Absorbing Chain

A particle performs a random walk on $\{0, 1, 2, 3, 4\}$ where:

- States 0 and 4 are absorbing
- From state 1: move to 0 with prob. 0.3, to 2 with prob. 0.7
- From state 2: move to 1, 3 each with prob. 0.5
- From state 3: move to 2 with prob. 0.6, to 4 with prob. 0.4

Questions:

1. Write the transition matrix in canonical form
2. Compute the fundamental matrix N
3. What's the probability of absorption at 0 starting from state 2?
4. Expected time to absorption from state 2?



14.4.1 Solution

Part 1: Transition Matrix in Canonical Form

Standard form: List transient states first (1, 2, 3), then absorbing states (0, 4).

State ordering: 1, 2, 3, 0, 4

Transition matrix:

$$P = \begin{pmatrix} 0 & 0.7 & 0 & 0.3 & 0 \\ 0.5 & 0 & 0.5 & 0 & 0 \\ 0 & 0.6 & 0 & 0 & 0.4 \\ 0 & 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 0 & 1 \end{pmatrix}$$

Canonical form: $P = \begin{pmatrix} Q & R \\ 0 & I \end{pmatrix}$

where:

- Q = transitions between transient states (3×3)
- R = transitions from transient to absorbing states (3×2)
- 0 = zero matrix (2×3)
- I = identity matrix for absorbing states (2×2)

$$Q = \begin{pmatrix} 0 & 0.7 & 0 \\ 0.5 & 0 & 0.5 \\ 0 & 0.6 & 0 \end{pmatrix} \quad R = \begin{pmatrix} 0.3 & 0 \\ 0 & 0 \\ 0 & 0.4 \end{pmatrix}$$

	To 1	To 2	To 3	To 0	To 4
From 1	0	0.7	0	0.3	0
From 2	0.5	0	0.5	0	0
From 3	0	0.6	0	0	0.4
— Q above —	— R above —				From 0

0	0	0	1	0	From 4
0	0	0	0	1	

Part 2: Fundamental Matrix N

The fundamental matrix is:

$$N = (I - Q)^{-1}$$

First, compute $I - Q$:

$$I - Q = \begin{pmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{pmatrix} - \begin{pmatrix} 0 & 0.7 & 0 \\ 0.5 & 0 & 0.5 \\ 0 & 0.6 & 0 \end{pmatrix} = \begin{pmatrix} 1 & -0.7 & 0 \\ -0.5 & 1 & -0.5 \\ 0 & -0.6 & 1 \end{pmatrix}$$

To find $(I - Q)^{-1}$, we use the formula for 3×3 matrix inverse or solve $(I - Q)N = I$.

Method: Solve system of equations

Let $N = \begin{pmatrix} n_{11} & n_{12} & n_{13} \\ n_{21} & n_{22} & n_{23} \\ n_{31} & n_{32} & n_{33} \end{pmatrix}$

From $(I - Q)N = I$:

$$\begin{pmatrix} 1 & -0.7 & 0 \\ -0.5 & 1 & -0.5 \\ 0 & -0.6 & 1 \end{pmatrix} \begin{pmatrix} n_{11} & n_{12} & n_{13} \\ n_{21} & n_{22} & n_{23} \\ n_{31} & n_{32} & n_{33} \end{pmatrix} = \begin{pmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{pmatrix}$$

Column 1 equations:

$$\begin{cases} n_{11} - 0.7n_{21} = 1 \\ -0.5n_{11} + n_{21} - 0.5n_{31} = 0 \\ -0.6n_{21} + n_{31} = 0 \end{cases}$$

From equation 3: $n_{31} = 0.6n_{21}$

Substitute into equation 2:

$$-0.5n_{11} + n_{21} - 0.5(0.6n_{21}) = 0$$

$$-0.5n_{11} + n_{21} - 0.3n_{21} = 0$$

$$-0.5n_{11} + 0.7n_{21} = 0$$

$$n_{21} = \frac{0.5}{0.7}n_{11} = \frac{5}{7}n_{11}$$

Substitute into equation 1:

$$n_{11} - 0.7 \cdot \frac{5}{7}n_{11} = 1$$

$$n_{11} - 0.5n_{11} = 1$$

$$0.5n_{11} = 1$$

$$n_{11} = 2$$

Therefore:

$$n_{21} = \frac{5}{7} \cdot 2 = \frac{10}{7} \approx 1.429$$

$$n_{31} = 0.6 \cdot \frac{10}{7} = \frac{6}{7} \approx 0.857$$

Column 2 equations:

$$\begin{cases} n_{12} - 0.7n_{22} = 0 \\ -0.5n_{12} + n_{22} - 0.5n_{32} = 1 \\ -0.6n_{22} + n_{32} = 0 \end{cases}$$

From equation 3: $n_{32} = 0.6n_{22}$ From equation 1: $n_{12} = 0.7n_{22}$

Substitute into equation 2:

$$-0.5(0.7n_{22}) + n_{22} - 0.5(0.6n_{22}) = 1$$

$$-0.35n_{22} + n_{22} - 0.3n_{22} = 1$$

$$0.35n_{22} = 1$$

$$n_{22} = \frac{1}{0.35} = \frac{20}{7} \approx 2.857$$

Therefore:

$$n_{12} = 0.7 \cdot \frac{20}{7} = 2$$

$$n_{32} = 0.6 \cdot \frac{20}{7} = \frac{12}{7} \approx 1.714$$

Column 3 equations:

$$\begin{cases} n_{13} - 0.7n_{23} = 0 \\ -0.5n_{13} + n_{23} - 0.5n_{33} = 0 \\ -0.6n_{23} + n_{33} = 1 \end{cases}$$

From equation 1: $n_{13} = 0.7n_{23}$

Substitute into equation 2:

$$-0.5(0.7n_{23}) + n_{23} - 0.5n_{33} = 0$$

$$-0.35n_{23} + n_{23} - 0.5n_{33} = 0$$

$$0.65n_{23} = 0.5n_{33}$$

$$n_{23} = \frac{0.5}{0.65}n_{33} = \frac{10}{13}n_{33}$$

Substitute into equation 3:

$$-0.6 \cdot \frac{10}{13}n_{33} + n_{33} = 1$$

$$-\frac{6}{13}n_{33} + n_{33} = 1$$

$$\frac{7}{13}n_{33} = 1$$

$$n_{33} = \frac{13}{7} \approx 1.857$$

Therefore:

$$n_{23} = \frac{10}{13} \cdot \frac{13}{7} = \frac{10}{7} \approx 1.429$$

$$n_{13} = 0.7 \cdot \frac{10}{7} = 1$$

Fundamental Matrix:

$$N = \begin{pmatrix} 2 & 2 & 1 \\ \frac{10}{7} & \frac{20}{7} & \frac{10}{7} \\ \frac{6}{7} & \frac{12}{7} & \frac{13}{7} \end{pmatrix} \approx \begin{pmatrix} 2.000 & 2.000 & 1.000 \\ 1.429 & 2.857 & 1.429 \\ 0.857 & 1.714 & 1.857 \end{pmatrix}$$

Interpretation of N :

- N_{ij} = expected number of times in transient state j starting from transient state i
- Example: $N_{21} = \frac{10}{7} \approx 1.43$ means starting from state 2, we expect to visit state 1 about 1.43 times before absorption

Start State	Visits to 1	Visits to 2	Visits to 3
1	2.000	2.000	1.000
2	1.429	2.857	1.429
3	0.857	1.714	1.857

Part 3: Probability of Absorption at State 0 from State 2

The absorption probability matrix is:

$$B = NR$$

$$B = \begin{pmatrix} 2 & 2 & 1 \\ \frac{10}{7} & \frac{20}{7} & \frac{10}{7} \\ \frac{6}{7} & \frac{12}{7} & \frac{13}{7} \end{pmatrix} \begin{pmatrix} 0.3 & 0 \\ 0 & 0 \\ 0 & 0.4 \end{pmatrix}$$

Computing B :

Row 2, Column 1 (start at 2, absorb at 0):

$$\begin{aligned} B_{21} &= n_{21} \cdot r_{10} + n_{22} \cdot r_{20} + n_{23} \cdot r_{30} \\ &= \left(\frac{10}{7}\right)(0.3) + \left(\frac{20}{7}\right)(0) + \left(\frac{10}{7}\right)(0) \\ &= \frac{3}{7} \approx 0.429 \end{aligned}$$

Row 2, Column 2 (start at 2, absorb at 4):

$$B_{22} = n_{21} \cdot r_{14} + n_{22} \cdot r_{24} + n_{23} \cdot r_{34}$$

$$\begin{aligned}
 &= \left(\frac{10}{7}\right)(0) + \left(\frac{20}{7}\right)(0) + \left(\frac{10}{7}\right)(0.4) \\
 &= \frac{4}{7} \approx 0.571
 \end{aligned}$$

Answer: Probability of absorption at state 0 starting from state 2 = $\frac{3}{7} \approx 0.429$ or 42.9%

Complete absorption probability matrix:

$$B = \begin{pmatrix} 0.6 & 0.4 \\ \frac{3}{7} & \frac{4}{7} \\ \frac{2}{7} & \frac{5}{7} \end{pmatrix} \approx \begin{pmatrix} 0.600 & 0.400 \\ 0.429 & 0.571 \\ 0.286 & 0.714 \end{pmatrix}$$

Start State	P(Absorb at 0)	P(Absorb at 4)	Verification
1	0.600 (60%)	0.400 (40%)	Sum = 1.0 ✓
2	0.429 (42.9%)	0.571 (57.1%)	Sum = 1.0 ✓
3	0.286 (28.6%)	0.714 (71.4%)	Sum = 1.0 ✓

Intuition: State 2 is equidistant from both absorbing states, but there's a slight bias toward state 4 due to the transition probabilities.

Part 4: Expected Time to Absorption from State 2

The expected time to absorption vector is:

$$t = N\mathbf{1}$$

where $\mathbf{1} = \begin{pmatrix} 1 \\ 1 \\ 1 \end{pmatrix}$ is a column vector of ones.

$$t = \begin{pmatrix} 2 & 2 & 1 \\ \frac{10}{7} & \frac{20}{7} & \frac{10}{7} \\ \frac{6}{7} & \frac{12}{7} & \frac{13}{7} \end{pmatrix} \begin{pmatrix} 1 \\ 1 \\ 1 \end{pmatrix}$$

For state 2 (row 2):

$$\begin{aligned}
 t_2 &= n_{21} \cdot 1 + n_{22} \cdot 1 + n_{23} \cdot 1 \\
 &= \frac{10}{7} + \frac{20}{7} + \frac{10}{7} \\
 &= \frac{40}{7} \approx 5.714 \text{ steps}
 \end{aligned}$$

Answer: Expected time to absorption from state 2 = $\frac{40}{7} \approx 5.71$ steps

Complete expected absorption times:

$$t = \begin{pmatrix} 2 + 2 + 1 \\ \frac{10}{7} + \frac{20}{7} + \frac{10}{7} \\ \frac{6}{7} + \frac{12}{7} + \frac{13}{7} \end{pmatrix} = \begin{pmatrix} 5 \\ \frac{40}{7} \\ \frac{31}{7} \end{pmatrix} \approx \begin{pmatrix} 5.000 \\ 5.714 \\ 4.429 \end{pmatrix}$$

Start State	Expected Steps to Absorption	Interpretation
-------------	------------------------------	----------------

1	5.000	Farthest from absorbing states
2	5.714	Central position, most visits
3	4.429	Closer to state 4

Verification through simulation reasoning: Starting from state 2:

- Visit state 2 itself 2.857 times (including start)
- Visit state 1 about 1.429 times
- Visit state 3 about 1.429 times
- Total visits: $2.857 + 1.429 + 1.429 = 5.715 \approx 40/7 \checkmark$

Summary for Problem 4

Property	Result
Transient States	1, 2, 3
Absorbing States	0, 4
Fundamental Matrix N	$\begin{pmatrix} 2 & 2 & 1 \\ \frac{10}{7} & \frac{20}{7} & \frac{10}{7} \\ \frac{6}{7} & \frac{12}{7} & \frac{13}{7} \end{pmatrix}$
$P(\text{Absorb at 0} \mid \text{Start at 2})$	$3/7 \approx 0.429$ (42.9%)
$P(\text{Absorb at 4} \mid \text{Start at 2})$	$4/7 \approx 0.571$ (57.1%)
Expected absorption time from 2	$40/7 \approx 5.71$ steps
Most likely outcome from 2	Absorption at state 4

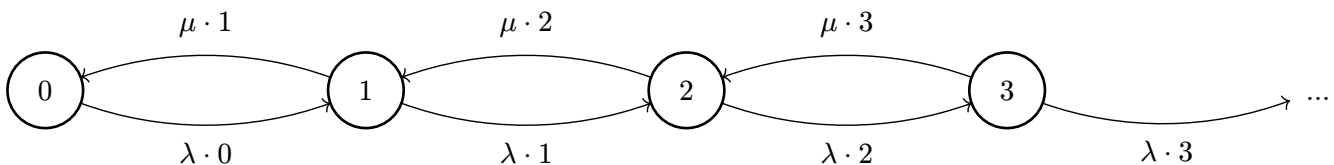
14.5 Problem 5: Birth-Death Process

A population follows a birth-death process with:

- Birth rate: $\lambda_n = \lambda n$ (proportional to population)
- Death rate: $\mu_n = \mu n$ (proportional to population)

Questions:

1. Write the transition rate matrix Q
2. Find the stationary distribution (if it exists)
3. Under what conditions is the process stable?



14.5.1 Solution

Part 1: Transition Rate Matrix Q

For a continuous-time Markov chain, the transition rate matrix Q has entries:

- q_{ij} = rate of transition from state i to state j (for $i \neq j$)
- $q_{ii} = -\sum_{j \neq i} q_{ij}$ (row sums to zero)

For the birth-death process:

- From state n : birth to $n + 1$ at rate $\lambda_n = \lambda n$
- From state n : death to $n - 1$ at rate $\mu_n = \mu n$

Special case: State 0 When population is 0, no births or deaths occur (extinction):

- $\lambda_0 = \lambda \cdot 0 = 0$ (no births from empty population)
- $\mu_0 = \mu \cdot 0 = 0$ (no deaths from empty population)
- State 0 is absorbing in this model!

For $n \geq 1$:

- $q_{n,n+1} = \lambda n$ (birth rate)
- $q_{n,n-1} = \mu n$ (death rate)
- $q_{nn} = -(\lambda n + \mu n)$ (total exit rate)

Transition rate matrix (infinite):

$$Q = \begin{pmatrix} 0 & 0 & 0 & 0 & 0 & \dots \\ \mu & -(\lambda + \mu) & \lambda & 0 & 0 & \dots \\ 0 & 2\mu & -2(\lambda + \mu) & 2\lambda & 0 & \dots \\ 0 & 0 & 3\mu & -3(\lambda + \mu) & 3\lambda & \dots \\ \vdots & \vdots & \vdots & \vdots & \vdots & \ddots \end{pmatrix}$$

Finite-state approximation (truncate at N):

For states $0, 1, 2, \dots, N$ (assuming no population above N):

$$Q = \begin{pmatrix} 0 & 0 & 0 & 0 & \dots \\ \mu & -(\lambda + \mu) & \lambda & 0 & \dots \\ 0 & 2\mu & -2(\lambda + \mu) & 2\lambda & \dots \\ \vdots & \vdots & \vdots & \vdots & \ddots \\ 0 & \dots & 0 & N\mu & -N\mu \end{pmatrix}$$

State n	Birth Rate	Death Rate	Exit Rate	Net Rate
0	0	0	0	0 (absorbing)
1	λ	μ	$\lambda + \mu$	$\lambda - \mu$
2	2λ	2μ	$2(\lambda + \mu)$	$2(\lambda - \mu)$
3	3λ	3μ	$3(\lambda + \mu)$	$3(\lambda - \mu)$
n	$n\lambda$	$n\mu$	$n(\lambda + \mu)$	$n(\lambda - \mu)$

Part 2: Stationary Distribution

For a continuous-time Markov chain, the stationary distribution π satisfies:

$$\pi Q = 0 \quad \text{and} \quad \sum_i \pi_i = 1$$

Detailed balance equations:

For a birth-death process, detailed balance means:

$$\pi_n \cdot q_{n,n+1} = \pi_{n+1} \cdot q_{n+1,n}$$

This gives:

$$\pi_n \cdot (\lambda n) = \pi_{n+1} \cdot (\mu(n+1))$$

$$\pi_{n+1} = \frac{\lambda n}{\mu(n+1)} \pi_n$$

Recursive solution:

Starting from π_0 :

$$\pi_1 = \frac{\lambda \cdot 0}{\mu \cdot 1} \pi_0 = 0$$

Since $\lambda_0 = 0$, we have $\pi_1 = 0$, which implies $\pi_n = 0$ for all $n \geq 1$.

This means the only stationary distribution is $\pi = [1, 0, 0, \dots]$ (all probability on state 0).

Interpretation: State 0 is absorbing! Once the population reaches 0, it stays there forever. Eventually, the population goes extinct with probability 1.

Answer: The stationary distribution is $\pi_0 = 1, \pi_n = 0$ for $n \geq 1$ (extinction is certain).

Modified Model: Constant Immigration

To avoid extinction, modify the model with constant immigration:

- Birth rate: $\lambda_n = \lambda n + \alpha$ (births + immigration)
- Death rate: $\mu_n = \mu n$

Now $\lambda_0 = \alpha > 0$, so state 0 is not absorbing.

For this model, detailed balance gives:

$$\pi_n(\lambda n + \alpha) = \pi_{n+1}\mu(n+1)$$

$$\pi_{n+1} = \frac{\lambda n + \alpha}{\mu(n+1)} \pi_n$$

This is more complex, but for $\lambda < \mu$, a proper stationary distribution exists.

Part 3: Stability Conditions

Case 1: Original Model (No Immigration)

The process is NOT stable in the traditional sense:

- State 0 is absorbing
- For any initial state $n > 0$, the population will eventually reach 0 with probability 1
- Time to extinction depends on $\frac{\lambda}{\mu}$ ratio

Mean time to extinction from state n :

Let T_n = expected time to reach 0 starting from n .

From the differential equations:

$$T_n = \frac{1}{n\mu} + \frac{\lambda}{\lambda + \mu} T_{n+1} + \frac{\mu}{\lambda + \mu} T_{n-1}$$

For $\lambda < \mu$ (deaths exceed births):

$$T_n \approx \frac{1}{n(\mu - \lambda)} \quad \text{for large } n$$

Extinction is faster when $\mu \gg \lambda$.

For $\lambda > \mu$ (births exceed deaths):

$$T_n \approx \exp\left(n \frac{\lambda - \mu}{\mu}\right)$$

Extinction takes exponentially long! But still happens eventually.

For $\lambda = \mu$ (balanced):

$$T_n \approx \frac{n^2}{2\mu}$$

Case 2: With Constant Immigration ($\lambda_0 = \alpha > 0$)

The process is stable if $\lambda < \mu$:

- **Stable ($\lambda < \mu$):** Unique stationary distribution exists

$$\pi_n \propto \left(\frac{\lambda}{\mu}\right)^n \cdot \prod_{k=0}^{n-1} \frac{\lambda k + \alpha}{\mu(k+1)}$$

- **Critical ($\lambda = \mu$):** Null recurrent or transient depending on α
- **Unstable ($\lambda > \mu$):** Population grows to infinity

$$E[X(t)] \rightarrow \infty \quad \text{as } t \rightarrow \infty$$

Stability condition summary:

Condition	Long-term Behavior	Stationary Distribution
$\lambda < \mu$	Population decreases to 0	Extinction (if no immigration)
$\lambda = \mu$	Random walk (neutral)	Extinction (slower)
$\lambda > \mu$	Population grows before extinction	Extinction (very slow)
With immigration, $\lambda < \mu$	Stable equilibrium	Proper distribution exists
With immigration, $\lambda \geq \mu$	Unbounded growth	No stationary distribution

Answer:

1. Original model: Always reaches extinction (state 0 absorbing)
2. With immigration: Stable if and only if $\lambda < \mu$ (death rate exceeds birth rate)

Summary for Problem 5

Property	Result
Type	Continuous-time birth-death process
Birth rate from n	$\lambda_n = \lambda n$
Death rate from n	$\mu_n = \mu n$
State 0	Absorbing (no births/deaths)
Stationary distribution	$\pi_0 = 1$, all others = 0
Long-term outcome	Extinction certain
Stability condition (with immigration)	$\lambda < \mu$ (deaths > births)
Key insight	Without external input, populations die out

Real-World Example: Population Dynamics

Consider a species with:

- Birth rate proportional to population: λn
- Death rate proportional to population: μn
- $\lambda = 0.8$ births per individual per year
- $\mu = 1.0$ deaths per individual per year

Since $\lambda < \mu$, population declines. Starting with 100 individuals:

- Expected time to extinction $\approx \frac{1}{100(1.0-0.8)} = \frac{1}{20} = 0.05$ years ≈ 18 days

This model explains why isolated populations without immigration eventually go extinct!

15 Summary and Key Takeaways

15.1 Essential Concepts

Concept	Key Point
Markov Property	Future depends only on present, not past
Transition Matrix	Stochastic matrix with rows summing to 1
Stationary Distribution	Satisfies $\pi P = \pi$; long-run behavior
Irreducibility	All states communicate
Aperiodicity	No cyclic behavior (period = 1)
Ergodicity	Aperiodic + positive recurrent
Absorbing State	$p_{ii} = 1$; can't escape

15.2 Important Theorems

1. **Ergodic Theorem:** For irreducible, aperiodic chains:

$$\lim_{n \rightarrow \infty} p_{ij}^{(n)} = \pi_j$$

2. **Fundamental Theorem:** Finite, irreducible, aperiodic chains have unique stationary distribution
3. **Detailed Balance:** Reversible chains satisfy $\pi_i p_{ij} = \pi_j p_{ji}$
4. **Chapman-Kolmogorov:** $p_{ij}^{(n+m)} = \sum_k p_{ik}^{(n)} p_{kj}^{(m)}$

15.3 Common Pitfalls to Avoid

1. **Confusing rows and columns** in transition matrices
 - Remember: rows represent current state, columns represent next state
 - Row sums must equal 1
2. **Assuming all chains have stationary distributions**
 - Reducible chains may not have unique stationary distribution
 - Null recurrent chains don't have stationary distribution
3. **Forgetting initial distribution**
 - P^n gives transition probabilities, not state distribution
 - State distribution: $\pi^{(n)} = \pi^{(0)} P^n$
4. **Misidentifying periodicity**
 - Self-loops guarantee aperiodicity
 - Check gcd of return times, not just presence of cycles

15.4 Study Tips

1. **Practice with examples:** Work through the problems systematically
2. **Visualize with diagrams:** Draw state transition diagrams
3. **Use software:** Implement algorithms in Python/R
4. **Connect to applications:** Understand real-world context
5. **Master fundamentals:** Linear algebra (eigenvalues, matrix powers)

16 References and Further Reading

16.1 Textbooks

1. **Norris, J.R.** (1997). *Markov Chains*. Cambridge University Press.
 - Comprehensive theoretical treatment
 - Graduate level
2. **Grinstead, C.M. & Snell, J.L.** (2012). *Introduction to Probability*.
 - Free online textbook
 - Accessible introduction with exercises
3. **Ross, S.M.** (2014). *Introduction to Probability Models* (11th ed.). Academic Press.
 - Excellent for applications
 - Clear explanations
4. **Kemeny, J.G. & Snell, J.L.** (1976). *Finite Markov Chains*. Springer.
 - Classic reference
 - Focus on finite chains

16.2 Online Resources

- MIT OpenCourseWare: Stochastic Processes
- Stanford CS109: Probability for Computer Scientists
- Brilliant.org: Interactive Markov Chain tutorials
- Wikipedia: Comprehensive reference with examples

16.3 Research Papers

1. Markov, A.A. (1906). “Extension of the limit theorems of probability theory to a sum of variables connected in a chain”
 - Original paper introducing Markov chains
2. Page, L. et al. (1999). “The PageRank Citation Ranking: Bringing Order to the Web”
 - Application to web search

16.4 Software Packages

- **Python:** NumPy, SciPy, PyMC3, hmmlearn
- **R:** markovchain, msm, depmixS4
- **MATLAB:** Statistics and Machine Learning Toolbox

17 Appendix: Mathematical Notation

Symbol	Meaning
X_n	State at time n
S	State space
p_{ij}	Transition probability from i to j
P	Transition matrix
$p_{ij}^{(n)}$	n -step transition probability
π	Stationary distribution
π_i	Stationary probability of state i
f_{ij}	Probability of ever reaching j from i
T_i	First return time to state i
$d(i)$	Period of state i
Q	Transition rate matrix (CTMC)
q_{ij}	Transition rate from i to j
N	Fundamental matrix (absorbing chains)
I	Identity matrix
$\mathbb{1}$	Vector of ones

End of Study Material

“The future is independent of the past, given the present.”
 — The Markov Property